Identify or Certify? What Public Market Data Reveal About Algorithmic Feedback

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Abstract

Regulators ask to know which algorithms are trading. We ask which questions about those algorithms market data can answer, and at what cost. In a closed-loop model in which cohorts of algorithms respond to prices with exponential memory, every question an outside observer can ask is a functional of the merged measure of the cohorts' memory rates and weights. Identifying the cohorts is expensive: separating K cohorts whose memory rates are $\Delta\eta$ apart needs a number of observations of order $(\Delta\eta)^{-(4K-2)}$, from experiments and from passive data alike. In the positive cohort class, certifying stability depends on the cohorts only through the stability margin $M = 1 - a - \kappa\Phi$, and from passive data deciding its sign takes a record of length of order $1/M$, with matching lower and upper bounds that hold for any number of cohorts with finite memory, uniformly as the margin vanishes, and, for fixed economies, for the price-and-flow estimator used in practice. Without a bound on the agents' memory no passive certificate exists: a slow momentum cohort can hide instability from any finite record. Tests registered before their data were downloaded, on disjoint sets of crypto perpetual futures never used before, support the dichotomy: the margin was decided, always on the safe side, in 22 of 30, 23 of 25 and 24 of 26 markets while the cohorts were identified in at most one per test, and on 34 further markets the margin forecast next-day variance beyond the Hawkes branching ratio. It did not forecast adverse or extreme moves and did not rank damage on the worst days. Under the maintained model and observation

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assumptions the estimates restrict a persistence margin; the empirical EMA proxy is not independently identified fundamental displacement. The tests support a reduced-form forecasting interpretation, not a general public-data safety certificate.

Research status update: 30 October 2026. This development revision records a verified returned synthetic experiment and a dated work register at the end. The new audit supports a conditional finite-horizon financial decision result, not general profitability or real-market safety. Earlier theoretical and historical claims are retained with their stated assumptions; this update does not independently verify every proof or rerun every earlier empirical study. Novelty, full proof review, economic information acquisition and the public-data bridge remain open.

Keywords: algorithmic trading, identification, certification, closed-loop markets, unit roots, disclosure, pre-registration. **JEL:** G14, G18, C12, C22.

1 Introduction

When trading algorithms condition on recent prices and their orders move prices, price formation is a feedback system. Supervisors have responded by asking to know which algorithms are trading. Under MiFID II, investment firms in the European Union must flag algorithmic orders and identify the algorithm responsible for each; the US Consolidated Audit Trail links every order in listed equities and options to an anonymized customer. The premise is that knowing the agents is the route to knowing whether the market is safe. This paper separates the two questions and shows that, for an observer who sees only market data, they have very different costs. The cheap one is the one that matters for stability.

We work with a linear closed-loop model. The displacement of the price from its fundamental value reverts at a known rate and is pushed by order flow with a known impact; K cohorts of algorithms track the displacement with exponential memory, the reduced form of steady-state Kalman filtering and of regularized online learning, and trade on it with weights that may be positive (momentum) or negative (contrarian). In this model the agents enter the observable dynamics only through a merged measure of memory rates and weights. Every question an outside observer can ask is a functional of that measure, and the questions differ sharply in cost.

Identification is expensive. Two cohorts whose memory rates differ by δ produce price responses that differ at order δ^2, so separating them needs $\Omega(\delta^{-4})$ observations; separating K cohorts spaced $\Delta\eta$ apart needs $\Omega((\Delta\eta)^{-(4K-2)})$. These bounds are attained, and they hold whether the observer runs controlled experiments or watches the market passively (Section 4).

Certification is cheap. Stability depends on the agents only through their total feedback weight Φ, and the closed loop is stable when the margin $M = 1 - a - \kappa\Phi$ is positive. Deciding the sign of M from a passive record is the problem of rejecting a unit root. We prove that it needs a record of length of order $1/M$, with matching lower and upper bounds, first for one closed-loop root (Theorem 5), then for any number of cohorts with finite memory (Theorem 6), uniformly as the margin vanishes (Theorem 10), and for the estimator that pools a price-impact regression with a flow regression, which is the estimator used in practice (Theorem 7). The number and similarity of the cohorts do not enter. Together with the identification bounds this gives the paper's central result (Theorem 8): the cost of identification is set by how similar the agents are, the cost of certification by how close the market is to instability, and for a market at a fixed margin the ratio of the two grows without bound as the agents become similar.

What disclosure must supply. Certification needs one thing that prices cannot provide: a bound on memory. We show that without it no passive certificate exists at all (Theorem 9). A momentum cohort that trades on long averages of the price contributes little to each period's response and a great deal to the total feedback, so it can make a market unstable without leaving a trace in any finite record; in simulation, a standard unit-root certificate certified such an unsafe market as safe in 100% of replications. The policy implication is specific. A supervisor who wants stability needs disclosures that bound how slowly the largest participants trade, not their identities.

What the margin measures. The margin decides stability. When some agents are contrarian it does not bound worst-case loss, because the impulse response can oscillate; the quantity that does is the horizon-limited absolute response, which we show is certified from passive data at the parametric rate (Theorem 12).

Evidence. We test these predictions on public Binance data with tests whose specifications, code hashes and universe rules were pushed to a public repository in commits of their own before any of their data were downloaded, each run once, and each on perpetual futures that appear in no earlier part of the project (Section 7). Three results stand out.

(i) *Certification without identification.* On three disjoint sets of new markets the margin was decided in 22 of 30 (one short of the registered threshold, with 105 days), 23 of 25 and 24 of 26 (with 430 days), always on the safe side, while the memories were identified in at most one market per test. Markets decided but not identified outnumbered the reverse by 22 to 1, 22 to 0 and 23 to 0.

(ii) *The certificate forecasts turbulence.* On 34 further markets, a lower margin forecast higher next-day realized variance out of sample (Clark–West $t = 4.82$), and it did so beyond the branching ratio of a Hawkes model of trade arrivals, the standard measure of self-excitation in markets ($t = 4.84$); the branching ratio alone did not ($t = -0.54$).

(iii) *It does not forecast crashes.* Registered forecasts of next-day adverse moves (F1) and extreme moves (S1) from the margin failed, and on the 20 worst market-wide days the prior margin did not rank which markets fell furthest. On 36 more markets, losses scaled with the estimated closed-loop response at about half the predicted rate, but beyond recent volatility a larger response forecast smaller losses, not larger.

Of the twenty-three registered primary hypotheses tested in the project, eleven passed and twelve failed; we report all of them (Table 1).

Contribution. The paper makes three contributions. The first is conceptual: it separates identification from certification as questions about the same closed-loop market, and shows that their costs are governed by different features of the market. The second is theoretical: a complete passive theory of certifying stability, with sharp rates, uniformity, the estimator used in practice, and an impossibility result that says exactly what disclosure must add. The third is empirical and methodological: a sequence of tests, each registered before its data existed in the analyst's hands, on markets never used before, whose failures are reported with the same weight as its successes.

What we do not claim. The bounds hold for the stated linear model and observation contracts. They are not limits on what participant-level records can reveal; with such records identification becomes ordinary estimation (Niroula and Niroula, 2026a). The empirical margin is estimated from a reduced form whose relation to the model rests on linearity, on the exogeneity of each period's flow to the period's price noise, on a moving-average proxy for the fundamental value, and on the stationarity of the margin over the estimation window; we say where each strains. Every certified market was far from the boundary, so the evidence shows that public data can certify stability, not that they detect a market approaching instability.

Section 2 places the paper in the literature. Section 3 sets out the model and the observation contracts. Section 4 states the identification bounds, Section 5 the certification theory, and Section 6 the results on memory, uniformity and worst-case loss. Section 7 reports the evidence, Section 8 exploratory robustness, and Section 9 the implications. Proofs not given in the text are in Appendix A; the registration record is in Appendix B.

2 Related literature

Price formation and feedback. The model is a closed-loop version of the linear price-impact tradition that starts with Kyle (1985). Order flow moves prices with an impact coefficient, and the information content of trades is measured by the dynamic response of prices to flow (Hasbrouck, 1991). The propagator view of Bouchaud et al. (2004) treats the price as a linear filter of past order flow and emphasizes that flow is itself persistent. In our model flow is persistent because algorithms respond to past prices, and the feedback loop is what makes stability a question at all. Brunnermeier and Pedersen (2009) show how funding constraints turn price moves into forced trades, and Kirilenko et al. (2017) document how high-frequency traders' inventory management amplified the 2010 Flash Crash. Kirilenko and Lo (2013) survey the supervisory problems algorithmic trading creates; Budish et al.

(2015) study a market design response to speed competition. We take the feedback loop as given and ask what an observer can learn about it.

Self-excitation. The closest empirical measure of feedback in markets is the branching ratio of a Hawkes model of order arrivals, which estimates the share of events triggered by earlier events (Filimonov and Sornette, 2012; Hardiman et al., 2013; Bacry et al., 2015). A branching ratio close to one has been read as a market close to criticality. The branching ratio describes the arrival of trades; the margin describes the price feedback loop, and the two can move independently. In a registered head-to-head comparison the margin forecast next-day variance and the branching ratio did not (Section 7.4).

Identification of mixtures. Recovering the cohorts from the merged measure is a problem of estimating a finite mixture from its moments, closely related to super-resolution. Rates of the form $(\Delta)^{-(4K-2)}$ for separating K closely spaced components appear in the theory of strong identifiability (Heinrich and Kahn, 2018), of moment methods (Wu and Yang, 2020), and of super-resolution under sparsity (Donoho, 1992). The companion paper (Niroula and Niroula, 2026a) proves the market form of these bounds and translates them into executed quantity and trading capital.

Linear functionals and certification. The cost of certifying a linear functional of an unknown measure from noisy linear observations is governed by a modulus of continuity (Ibragimov and Has'minskii, 1984; Donoho, 1994). What is specific to markets is that the stability functional, the total feedback weight, sits exactly on the boundary of the class that can be certified at the parametric rate, and pays one logarithm (Niroula, 2026b).

Unit roots. Passive certification is the problem of rejecting a unit root. We use the limit theory of Mann and Wald (1943), Phillips (1987), Chan and Wei (1987) and Phillips and Magdalinos (2007), the uniform theory of Mikusheva (2007, 2012), and the augmented regression of Dickey and Fuller (1979) and Said and Dickey (1984). That unit-root tests have little power when the root is near one (Elliott et al., 1996), and no power against all stationary alternatives (Blough, 1992; Faust, 1996; Pötscher, 2002), are classical. Our contributions here are the market form: the dependence of the cost on the margin and on the root structure that the agents' memories create, the pooled estimator, and the identification of slow momentum cohorts as the economic content of the impossibility.

Credibility of empirical finance. Harvey et al. (2016) and Harvey (2017) argue that the multiplicity of tests in finance demands higher evidentiary standards and pre-specification.

Our tests follow that advice literally: each specification was made public before its data were downloaded, each test was run once, and the results, including twelve failed primaries, are reported as they came out.

3 Model and observation contracts

3.1 The closed-loop economy

The displacement of the price from its fundamental value, x_t, evolves as

$$x_{t+1} = a\,x_t + \kappa\,(u_t + w_{t+1}), \qquad u_t = \sum_{i=1}^{K} q_i\,y_{i,t}, \qquad y_{i,t} = r_i\,y_{i,t-1} + (1-r_i)\,x_t, \tag{1}$$

with persistence $a \in [0,1)$, impact $\kappa > 0$, exogenous order-flow shocks w_t, and K cohorts with weights $q_i \in \mathbb{R}$ and memory rates $r_i \in [0,1)$. The state $y_{i,t}$ is cohort i's estimate of the displacement; exponential smoothing is the steady state of a Kalman filter for a persistent signal in noise and the fixed point of online gradient descent with L_2 regularization (Niroula and Niroula, 2026a). A cohort with $q_i > 0$ submits positive demand when its smoothed displacement is positive; this is a reinforcing response assumption, not a derived fundamental-value trading optimum. A negative weight gives a contrarian response.

The aggregate response is a linear filter of past displacements,

$$u_t = \sum_{m\geq 0} k_m\,x_{t-m}, \qquad k_m = \int r^m\,dQ(r), \qquad Q = \sum_{i=1}^{K} q_i(1-r_i)\,\delta_{r_i},$$

so the agents enter the observable dynamics only through the *merged measure* Q: cohorts that share a memory rate are indistinguishable whatever their identities. The total feedback weight is $\Phi = \sum_m k_m = \int (1-r)^{-1} dQ(r) = \sum_i q_i$, and the closed-loop polynomial $f(z) = 1 - az - \kappa z \sum_m k_m z^m$ has $f(1) = 1 - a - \kappa\Phi$.

Definition 1 (Stability margin and summed response)**.** The *stability margin* is $M = 1 - a - \kappa\Phi = f(1)$. The *impulse response* h_t is the effect on x_{t-1} of a unit flow shock at time zero; the *summed response* is $S = \sum_t h_t = \kappa/M$ when the sum converges.

In the positive class (every $q_i \geq 0$), every tap of h is non-negative, the closed loop is stable if and only if $M > 0$, and the worst-case displacement caused by shocks bounded by $\bar{w}$ is $\bar{w}\|h\|_1 = \bar{w}S$. Stability, and in the positive class worst-case loss, are therefore questions about one linear functional of Q, while identification asks for Q itself. With signed weights

a stable loop can have an oscillating response, $\|h\|_1$ can exceed $|S|$ by an unbounded factor, and stability and worst-case loss come apart (Section 6.3).

3.2 Observation contracts

We compare questions under two observation contracts. Under *reset experiments*, an observer can restart the market from rest, apply a probe and read the noisy response; this requires an independently justified experimental or audited-reset setting. Ordinary market access or the ability to trade does not provide a market-wide reset. Under *passive data*, the observer only watches the market run and records prices and, when available, anonymous signed order flow; this is the contract of anyone with public data. A third contract, *tagged disclosure*, gives the observer each class's own flow or positions; it turns identification into ordinary estimation and is the subject of the companion paper.

3.3 From the model to the estimator

The tests estimate two regressions per market and pool their sufficient statistics over days. With x_t the log price minus its exponential moving average on five-minute marks and f_t signed taker notional,

$$x_{t+1} - x_t = \kappa f_{t+1} + \beta x_t + \varepsilon_{t+1}, \qquad f_{t+1} = \sum_{m<L} k_m x_{t-m} + w_{t+1}, \tag{2}$$

with $L = 12$ lags, so that $\beta = a - 1$ and the margin is $M = -\beta - \kappa\Phi$. The mapping from (1) to (2) needs three assumptions, each of which can fail. The kernel must be well approximated by twelve lags (one hour); Theorem 9 shows what is lost when it is not. The price noise ε_{t+1} must be independent of the period's flow, the usual identifying assumption of impact regressions; in five-minute bars the two can be simultaneous. And the moving-average displacement must stand in for the distance from fundamental value; when the fundamental moves, the proxy confounds reversion with tracking.

4 Identification is expensive

Theorem 2 (Merged equivalence and the cost of separation; Niroula and Niroula, 2026a)**.** *Let a and κ be known.*

(i) Two economies produce identical price responses to every probe if and only if their merged measures coincide.

(ii) With reset experiments whose noisy responses have variance σ^2, separating a single cohort at memory rate $\bar{r}$ from two cohorts at $\bar{r} \pm \delta$ with the same total weight needs $\Omega(\sigma^2\delta^{-4})$ experiments under an average-power constraint on probes; separating two K-cohort economies whose rates lie in an interval of width $2\Delta\eta$ and whose merged measures share their first $2K-1$ moments needs $N \geq c\,\sigma^2(\Delta\eta)^{-(4K-2)}$.

(iii) With passive data, the Kullback–Leibler divergence rate between the same pairs is $\Theta(\delta^4)$ and $\Theta((\Delta\eta)^{4K-2})$, so a passive record needs length $\Omega(\delta^{-4})$ and $\Omega((\Delta\eta)^{-(4K-2)})$, and the likelihood-ratio test attains these orders.

Part (i) is the reason identification is hard. The response to a probe depends on the moments $k_m = \int r^m dQ$, and two measures with the same first moments produce responses that agree to high order; part (ii) quantifies how high. The companion paper translates the bounds into the executed quantity, of order δ^{-2}, and trading capital, of order δ^{-4}, that a probing supervisor would need for a known pair, and shows that settled online learners trained in the closed loop obey the same scaling. The point for this paper is part (iii): watching instead of probing does not make identification cheaper, and nothing in the bounds depends on how close the market is to instability.

5 Certification is cheap

5.1 Reset experiments

Under the reset contract, a linear question $F_g(Q) = \int g\,dQ$ over measures on $[0, \bar{r}]$ with mass at most B costs a number of experiments set by a modulus of continuity whose dual is a problem of polynomial approximation with penalized coefficients; it is parametric exactly when g has square-summable power-series coefficients (Niroula, 2026b, Chapter 4). The total feedback Φ has $g(r) = 1/(1-r)$, whose coefficients are all one: it sits on the boundary.

Theorem 3 (The stability question costs one logarithm; Niroula, 2026b)**.** *Under the reset contract, with memory bound $\bar{r} < 1$, certifying Φ, and S whenever $a + \kappa B/(1-\bar{r}) < 1$, to half-width ϵ needs $N \asymp \sigma^2\epsilon^{-2}\log(1/\epsilon)$ experiments as $\epsilon \to 0$, and the logarithm cannot be removed. Without a memory bound, no finite number of experiments certifies Φ to any finite accuracy.*

Corollary 4 (What the closed loop adds)**.** *In the positive class with $M > 0$ and $\epsilon \in (0, 1/2]$, certifying S to relative accuracy ϵ is equivalent, up to a factor of two in accuracy, to certifying Φ to half-width $\epsilon M/\kappa$; it costs of order $\sigma^2\kappa^2 M^{-2}\epsilon^{-2}\log(\kappa/(\epsilon M))$ experiments.*

Proof. S is increasing in Φ on $\Phi < (1-a)/\kappa$. If $|\Phi' - \Phi| \le \epsilon M/(2\kappa)$ then $M' = 1 - a - \kappa\Phi' \ge M(1 - \epsilon/2)$ and $|S' - S|/S = \kappa|\Phi' - \Phi|/M' \le \epsilon$. Conversely $|S' - S| \le \epsilon S$ gives $\kappa|\Phi' - \Phi| \le \epsilon M' \le \epsilon(M + \kappa|\Phi' - \Phi|)$, so $\kappa|\Phi' - \Phi| \le 2\epsilon M$. Substitute into Theorem 3. □

The corollary answers the objection that the closed loop is a reparametrization of a known moment problem. For identification it is. For stability it is not: the loop turns a linear question about Φ into a question about S whose cost grows as the market nears the boundary. The economics of the problem, how close the market is to instability, enters the cost of certification and only of certification.

5.2 Passive data: one root

Theorem 5 (Passive certification, one root)**.** *Observe $x_{t+1} = a\,x_t + \kappa \sum_{m\ge 0} k_m x_{t-m} + \kappa w_{t+1}$ with independent $N(0, s^2)$ shocks for $t < T$, from $x_0 = 0$.*

(i) For $\mu \in (0, 1]$, two economies with one cohort of memory rate zero and margins μ and $-\mu$ have Kullback–Leibler divergence at most $2T\mu$; deciding the sign with error at most $\alpha < 1/2$ under both needs $T \ge (1 - 2\alpha)^2/\mu$.

(ii) In these economies $x_{t+1} = \rho x_t + \kappa w_{t+1}$ with $\rho = 1 - M$. For fixed $\rho \in (-1, 1)$, $\sqrt{T}(\hat{M} - M) \to N(0, M(2 - M))$ for the least-squares $\hat{M} = 1 - \hat{\rho}$; along $M_T \to 0$ with $TM_T \to \infty$, $(T/2M_T)^{1/2}(\hat{M} - M_T) \to N(0, 1)$. The studentized interval decides the sign with asymptotic error at most α once $TM > 2z^2_{1-\alpha/2}$.

Part (i) is the reason unit-root tests have little power when $T(1-\rho)$ is small (Elliott et al., 1996): a market close to the boundary cannot be certified from a short record, whoever its agents are.

5.3 Passive data: several cohorts

Theorem 6 (Passive certification, several cohorts)**.** *Let the kernel vanish beyond $L \ge 2$ lags, with any signs, and write the economy as $x_{t+1} = \sum_{i<L} \phi_i x_{t-i} + e_{t+1}$ with e_t independent $N(0, \sigma^2)$ and $x_t = 0$ for $t \le 0$, so that $f(z) = 1 - \sum_{i<L} \phi_i z^{i+1}$ and $M = f(1)$. Suppose $f(z) = (1-\rho z)g(z)$ with g real, $g(0) = 1$, of degree $p = L-1$ and without zeros in $|z| \le 1+\gamma$, $\gamma > 0$. Then $g(1) > 0$, $M = (1 - \rho)g(1)$, and:*

(i) for $0 < \mu \le g(1)$ the economies with $\rho_\pm = 1 \mp \mu/g(1)$ have margins $\pm\mu$ and divergence at most $2T\mu/g(1)$; deciding the sign with error at most $\alpha < 1/2$ needs $T \ge (1 - 2\alpha)^2 g(1)/\mu$;

(ii) for $\rho \in [0,1)$, the least-squares estimate of $-M$ in the exact augmented regression of $x_{t+1} - x_t$ on x_t and p lagged differences satisfies $\sqrt{T}(\hat{M} - M) \to N(0, V)$ with

$$V \le C_{p,\gamma} M, \qquad C_{p,\gamma} = 4p^2(p+1)(1+\pi^2)\, 4^p \big((1+\gamma)/\gamma\big)^p,$$

so the studentized interval has half-width below M once $T > z^2_{1-\alpha/2} C_{p,\gamma}/M$.

The constant in part (ii) is crude. In numerical checks with $p = 2$ and ρ up to 0.99, V/M is within 15% of $2g(1)$, so that the upper bound needs T of order $2z^2 g(1)/M$ and the lower bound $(1-2\alpha)^2 g(1)/M$: they match in their dependence on both $g(1)$ and M. The key step of the proof is a lower bound on the variance of the displacement that remains after projecting on its recent changes: whatever polynomial in the lag operator is applied, its value at frequency zero is one, and the spectrum of the displacement concentrates there like $1/\delta$, $\delta = 1 - \rho$.

5.4 The estimator used in practice

The estimator of the tests pools the price-impact regression and the flow regression in (2). It certifies at the same order.

Theorem 7 (The pooled price-and-flow estimator)**.** *Observe (x_t, f_t) generated by (2) with finite memory, $w_t \sim N(0, s_w^2)$ and $\varepsilon_t \sim N(0, s_\varepsilon^2)$ independent of each other and over time, $\kappa > 0$, $x_t = 0$ for $t \le 0$, and f as in Theorem 6. Let $\hat{M} = -\hat{\beta} - \hat{\kappa} \sum_m \hat{k}_m$ with $(\hat{\kappa}, \hat{\beta})$ and $\hat{k}$ the least-squares estimates of the two equations.*

(i) For $\mu \in (0,1]$, two economies with one cohort of memory rate zero, the same a and κ, and margins $\pm\mu$ have divergence at most $2T\mu(1+R)$ over T observations of (x_t, f_t), with $R = s_\varepsilon^2/(\kappa^2 s_w^2)$; deciding the sign with error at most $\alpha < 1/2$ needs $T \ge (1-2\alpha)^2/(\mu(1+R))$.

(ii) For $\rho \in [0,1)$, $\sqrt{T}(\hat{M} - M) \to N(0, V_{\text{pool}})$ with $V_{\text{pool}} \le 4C_{p,\gamma} M$.

Observing the flow changes the constant, not the order. The proof (Appendix A) writes $\hat{M} - M$ as the error of a price-equation coefficient on the level of the displacement plus κ times the error of the summed flow kernel, and shows that each is the coefficient on the level in a regression whose other regressors are stationary, so each inherits the variance bound of Theorem 6. In simulation $T\,\mathrm{Var}(\hat{M})/M$ was between 1.36 and 1.46 as ρ ranged from 0.9 to 0.99.

5.5 The dichotomy

Theorem 8 (Identify or certify)**.** *Under the passive contract, separating K cohorts whose memory rates are $\Delta\eta$ apart needs a record of length $\Omega((\Delta\eta)^{-(4K-2)})$ whatever the margin, while deciding the stability of a market with finite memory at margin M needs a record of length of order $1/M$ whatever the cohorts' spacing, with constants depending only on the memory length and the root gap, and the same order holds for the pooled estimator. For a market at a fixed margin, the ratio of the two costs grows without bound as the cohorts become similar.*

Proof. Theorem 2(iii) and Theorems 6 and 7; the latter's constants do not involve the cohorts' memory rates beyond the root gap of g, and their bounds do not involve $\Delta\eta$. □

5.6 A drifting margin

Estimated margins drift over months while their sign stays put. If each day gives an unbiased reading $m_t = M_t + e_t$ with noise $N(0, s^2)$ and the margin moves by at most d per day, averaging the last W readings has bias at most $d(W+1)/2$ and standard deviation $s/\sqrt{W}$; the best window is $W^* = (zs/d)^{2/3}$, and no procedure, using any history, can certify margins below a floor of order $(s^2 d)^{1/3}$ (Niroula, 2026b, Proposition 4.7.4). Calibrated to quarter-by-quarter estimates on 27 spent markets, the best window is about 53 days, close to one quarter, and the floor lies between 0.007 and 0.075.

6 Memory, uniformity, and what the margin measures

6.1 The price of memory

Theorem 9 (Hidden slow feedback)**.** *Let $\mathcal{E}_0$ be the passive economy with one cohort of memory rate zero and margin $c \in (0, 1]$, and let $\mathcal{E}_r$ add a momentum cohort with memory rate $r < 1$ and weight $(c+\mu)/\kappa$, so that its margin is $-\mu < 0$. Over T observations from rest, the divergence between them is at most $(c+\mu)^2(1-r)^2T^3/(2c)$. Hence, for every T and every procedure, $\sup_{r<1} \Pr_{\mathcal{E}_r}(\text{certify safe}) \geq \Pr_{\mathcal{E}_0}(\text{certify safe})$: without a memory bound, passive certification has power at most its size, however safe the market. With memory bound $\bar{r}$, deciding the sign with error at most α needs $T \geq \big(2(1-2\alpha)\sqrt{c}/((c+\mu)(1-\bar{r}))\big)^{2/3}$.*

Proof. The conditional means differ by $D_t = (c+\mu)(1-r)\sum_{m\leq t} r^m x_{t-m}$; by Cauchy–Schwarz and $\mathbb{E}_0 x_s^2 \leq \sigma^2/c$, $\mathbb{E}_0 D_t^2 \leq (c+\mu)^2(1-r)^2(t+1)^2\sigma^2/c$, and summing over $t < T$ gives the bound. Pinsker's inequality gives the rest. □

The theorem is the market form of the observation that unit-root tests have power no greater than size against all stationary processes (Blough, 1992; Faust, 1996; Pötscher, 2002). Its economic content is specific. In simulation with $c = \mu = 0.2$ and $T = 1000$, an augmented Dickey–Fuller certificate certified the safe economy in 100% of replications and the unsafe economy with memory rate 0.999 in 100%. The same argument applies when the flow is observed, because the slow cohort's contribution to each period's flow is of order $1 - r$. What disclosure must supply for stability is therefore a bound on how slowly the largest participants trade.

6.2 Uniformity

Theorem 10 (Uniform certification with finite memory)**.** *On the finite-memory economies of Theorem 6 with $\rho \in [0, 1]$ and g in a compact set with root gap γ, the augmented Dickey–Fuller certificate at level α has size tending to α uniformly on the boundary $M = 0$, tends to zero for every fixed explosive economy, and for every $\epsilon > 0$ certifies stability with probability at least $1 - \epsilon$, uniformly, once $TM/g(1) \geq C_\epsilon$. With Theorem 6(i), the record length needed is of order $g(1)/M$ uniformly as $M \to 0$.*

Proof. With $\theta = T(1 - \rho) = TM/g(1)$, the t-statistic converges to the local-to-unity functional τ_θ of an Ornstein–Uhlenbeck process (Phillips, 1987; Chan and Wei, 1987), uniformly in $\theta \geq 0$ and in the nuisance (Mikusheva, 2007, 2012); τ_0 is the Dickey–Fuller law and $\tau_\theta \to -\infty$ as $\theta \to \infty$. □

In simulation (Figure 1) the certification rate depended on the margin only through $TM/g(1)$: across two choices of g and $T \in \{500, 2000\}$ it was 4–7% at $\theta = 0$, 68–78% at $\theta = 10$ and at least 100% at $\theta = 40$.

6.3 Contrarian agents and worst-case loss

Proposition 11 (Contrarian agents)**.** *Allow signed weights, so that the merged measure is signed with total variation at most B.*

(i) Linear questions, including Φ, cost at most a constant factor more than in the positive class: the modulus of continuity at most doubles.

(ii) The summed response no longer measures risk. The worst-case loss is $|I|\bar{w}\|h\|_1 \geq |I|\bar{w}|S|$, and $\|h\|_1/|S|$ is unbounded on stable economies with a memory bound: one contrarian cohort with memory rate zero and $a + \kappa q = -q_0$, $q_0 \in [0, 1)$, has $h_t = \kappa(-q_0)^{t-1}$, $|S| = \kappa/(1 + q_0)$ and $\|h\|_1 = \kappa/(1 - q_0)$.

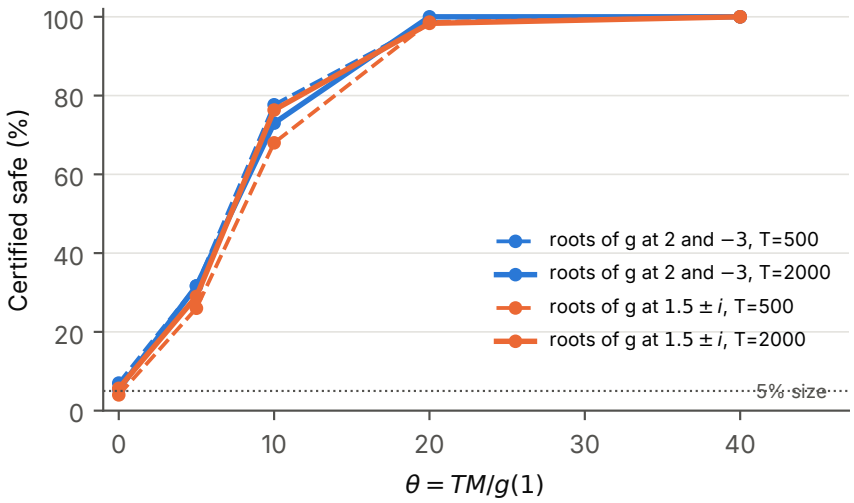


Figure 1: Simulated success of the augmented Dickey–Fuller certificate at the 5% level as a function of $\theta = TM/g(1)$, for two root structures of g and two record lengths. The curves collapse onto one, as Theorem 10 says: the margin matters only through $TM/g(1)$. 300 replications per point.

(iii) The horizon-limited worst case $\bar{w}\|h\|_{1,H}$ is certified from reset experiments with no memory bound and no sign condition, at a cost of order $\sigma^2 H^2 \epsilon^{-2}$ instead of $\sigma^2 H \epsilon^{-2}$ in the positive class.

Proof. (ii) With memory rate zero, $k_0 = q$ and $h_{t+1} = (a + \kappa q)h_t$ with $h_1 = \kappa$; sum the geometric series. Parts (i) and (iii) are proved in Niroula (2026b, Proposition 4.9.1). □

With ψ_j the coefficients of $1/f$, a shock sequence of k innovation standard deviations moves the displacement by at most $k\sigma\Psi_H$ within H periods, $\Psi_H = \sum_{j<H} |\psi_j|$, and some sequence attains it.

Theorem 12 (Passive certificate of worst-case loss)**.** *On a compact set of stationary finite-memory economies with signed kernels, the interval $\Psi_H(\hat{\phi}) \pm G_H(\chi^2_{L,1-\alpha}\lambda_{\max}(\hat{\Sigma})/T)^{1/2}$, with G_H a Lipschitz constant of Ψ_H and $\hat{\Sigma}$ the estimated covariance of the least-squares coefficients, covers Ψ_H with probability at least $1 - \alpha + o(1)$ uniformly; its half-width is of order $T^{-1/2}$, and no interval does better than order $T^{-1/2}$. The summed response $1/M$ can understate Ψ_∞ by an unbounded factor.*

Proof. Uniform asymptotic normality of $\hat{\phi}$ on compact sets of stationary autoregressions (Mann and Wald, 1943; Mikusheva, 2007), the polynomial dependence of each ψ_j on ϕ, and $\big||u| - |v|\big| \le |u - v|$ give the upper bound. For the lower bound, $\Psi_H = \sum_{j<H} \rho^j$ in the

memory-zero economies is strictly increasing with derivative at least one, and roots ρ and $\rho + \delta T^{-1/2}$ have divergence bounded in T. □

In a simulated contrarian economy the interval covered Ψ_H in 100% of replications, and the summed response understated it by 38%. The margin is the right statistic for whether a market is stable, in either class; it is the right statistic for worst-case loss only in the positive class.

7 Evidence

7.1 Data and design

All tests use Binance USD-M perpetual futures from the exchange's public archive (`data.binance.vision`): one-minute klines with taker-buy volume, which give the last price and the signed taker notional of each minute, aggregated to five-minute marks. A day is valid if at least half of its five-minute returns are non-zero and its flow is not identically zero; a market is evaluable with at least 200 valid days. On each valid day the sufficient statistics of the two regressions in (2) are computed and pooled over days, and the margin is $\hat{M} = -\hat{\beta} - \hat{\kappa}\hat{\Phi}$, decided if its 95% day-bootstrap interval (200 draws) excludes zero. The memories are identified if a two-exponential fit to the return kernel separates two rates with a bootstrap interval excluding zero and neither rate's interval touching a bound of the fit.

Each test's specification, the hashes of its code, its universe rule and its seed were pushed to a public repository in a commit of their own before any of its data were downloaded (Appendix B). The code refuses to run if any hash differs or the specification is not committed, and a counter that the code increments on opening prevents a second run. Universes are drawn in a seeded random order from symbols that appear in no file of the repository, so every test uses markets never used before. Results are reported as they came out, and explanations found afterwards are labelled post hoc. Table 1 lists every registered primary hypothesis of the project and its result.

7.2 Certification without identification

Table 2 and Figure 2 give the result. With 105 days per market the margin was decided in 22 of 30, one short of the threshold; with 430 days, in 23 of 25 and 24 of 26, and on the first 105 days of C3's markets in 20: the decision comes with the length of the record, as Theorem 5 predicts. Every decided margin was on the safe side, and the memories were

Table 1: Every registered primary hypothesis and its result

Test	Primary hypothesis	Result
E1	1. Certifying the summed response is at least 10 times more precise than separating memories	failed: 10 of 30 symbols (23 needed)
	2. Price-only memory intervals are at least 10 times wider than class-tagged ones	passed: 30 of 30
D1	1. Class-tagged interval width falls with slope in $[-0.75, -0.25]$	failed: 20 of 29 (22 needed)
	2. The advantage of tagging grows with the sample	passed: 22 of 29
G1	1. Amplification-adjusted volatility limit beats a volatility limit in stress hours	failed: identical rules ($\theta = 0$)
	2. The summed-response guard beats a volatility limit in stress hours	failed: tail-loss ratio +4.5% higher
V1	1. The depth-free statistic forecasts excursions in thin Kraken pairs	failed: $\Delta R^2 = -0.05$ points
	2. It beats an order-splitting placebo	not tested
O1	After scheduled US macro releases, small perpetuals overshoot	failed: $\bar{A}_{120} = 27.4$ s, interval [11,39]
C2	1. The margin is decided in at least 75% of 30 new markets	failed: 22 of 30 (23 needed)
	2. Decided but not identified outnumbers identified but not decided	passed: 22 against 1
	3. At a matched horizon, class-tagged persistence is tighter than price-only memory	passed: 29 of 29 (22 needed)
C3	1. With 430 days, the margin is decided in at least 75% of new markets	passed: 23 of 25 (19 needed)
	2. Decided but not identified outnumbers the reverse	passed: 22 against 0
F1	1. The margin is decided in at least 75% of new markets	passed: 24 of 26 (20 needed)
	2. Decided but not identified outnumbers the reverse	passed: 23 against 0
	3. The 20-day margin forecasts the next day's largest adverse move	failed: $t = 1.51$, $p = 0.065$
S1	1. On stress days, markets with a lower prior margin are hit harder	failed: mean partial correlation 0.000
	2. A lower margin forecasts next-day realized variance	passed: $t = 4.82$
	3. It does so beyond the Hawkes branching ratio	passed: $t = 4.84$
	4. A lower margin forecasts next-day extreme moves	failed: $t = 1.54$, Holm $p = 0.12$
R1	1. Next-day adverse moves rise with the absolute response Ψ_H	passed: elasticity 0.59 (0.09)
	2. Ψ_H forecasts adverse moves, positive coefficient	failed: coefficient -0.037
	3. It does so beyond the margin	failed: coefficient -0.040
H1	1. Real accounts have distinct memories their records reveal	prospective; opens 10 December 2026
	2. Prices and anonymous flow do not reveal them	prospective; opens 10 December 2026

Holm's correction across the primaries of each test (a fixed sequence for V1). E1, D1 and the first part of the project used other designs, reported in the companion thesis. Sources: the committed result files of each test, checked against their registered specification hashes by the project's verification script.

Table 2: Registered tests of the dichotomy

	C2	C3	F1
New markets drawn / evaluable	30 / 30	30 / 25	30 / 26
Days per market	105	430	430
Margin decided (registered threshold)	22 (23)	23 (19)	24 (20)
of which unsafe	0	0	0
Memories identified	1	1	1
Decided, not identified : identified, not decided	22 : 1	22 : 0	23 : 0
Same decision without the feedback term	26	22	22
Feedback term negative (contrarian)	14	16	14

C2 used 105 dates; C3 and F1 used 430 dates from January 2025 to September 2026 and C3's valid-day rule. Markets with fewer than 200 valid days were excluded under that rule.

identified in at most one market per test. Quarter by quarter, in an exploratory analysis of 169 symbol-quarters, the margin was decided in 135, never on the unsafe side.

Two qualifications matter. The decision rests mainly on reversion: without the feedback term it was the same in most markets, so these data certify that the markets are far from the boundary without measuring the feedback precisely. And the prediction that the standardized margin grows like the square root of the record holds across markets but not market by market, because the margin drifts; the drift calibration above puts most of these markets between the two ends of the drift floor.

7.3 Disclosure

Binance publishes five-minute long/short positioning ratios for a top-trader class and for all accounts. In C2 a registered primary compared the width of the interval for the class persistence rate with that of the price-only slow memory from a kernel matched to the class horizon (eight hours). The class interval was narrower in 29 of 29 markets, by a median factor of 7.3 (smallest 2.6). A registered secondary qualifies what this measures: corrected for the downward bias of day-demeaned regressions, the median class persistence was 1.0008 per five minutes, and below one in only 7 of 87 class fits. Positioning data restrict a persistence parameter that prices cannot bound; whether it reflects learning memory or slow turnover of positions, these data do not say. Test H1 below is designed to answer that with participant-level data.

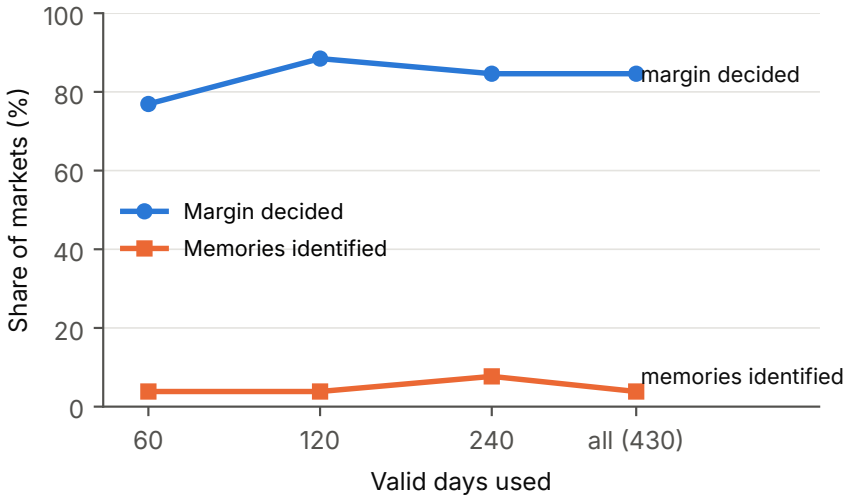


Figure 2: Test F1, registered secondary: share of the 26 evaluable new markets in which the margin was decided and in which the memories were identified, using the first 60, 120 and 240 valid days and all of them.

7.4 The certificate and risk

F1. A third registered primary of F1 asked whether the margin estimated on the previous 20 valid days forecasts the next valid day's largest adverse move out of sample, beyond realized variance, in a pooled model with symbol fixed effects and Clark–West loss differentials averaged within dates. The coefficient had the predicted sign (-1.13), but the test failed ($t = 1.51$, $p = 0.065$). For next-day realized variance, a registered secondary, the same design gave $t = 2.92$.

S1. Test S1 was registered on 25 October 2026 on 40 further markets and every day from January 2025 to September 2026 (34 evaluable). Two of its four primaries passed. A lower margin forecast higher next-day realized variance out of sample (coefficient -1.57, Clark–West $t = 4.82$, Holm-adjusted $p = 3 \times 10^{-6}$, gain 0.64 points of R^2 on 444 test dates), and the forecast was unchanged when the branching ratio of a Hawkes model of trade arrivals, estimated by the INAR approximation of Kirchner (2016), was added to both models ($t = 4.84$); the branching ratio alone did not forecast variance ($t = -0.54$). Two failed. On the 20 worst market-wide days the prior margin did not rank which markets were hit hardest (mean partial rank correlation 0.000), and the forecast of next-day extreme moves fell short ($t = 1.54$, Holm $p = 0.12$). As a secondary, the branching ratio did rank stress-day damage, but with the sign opposite to the criticality reading: markets with a higher prior branching ratio had smaller adverse moves (mean partial correlation -0.11).

R1. Test R1, registered the same day on 40 further markets (36 evaluable, none shared with S1), asked whether the estimated absolute response Ψ_H of Theorem 12, with H one hour, does what the margin did not. Within markets the next day's log adverse move rose with log Ψ_H, instrumented by its lag and controlling for shock size: elasticity 0.59 (standard error 0.09), positive as predicted, about half the theory's value of one; for variance the elasticity was 1.07 against two. Out of sample and beyond recent volatility, however, a larger Ψ_H forecast a smaller adverse move (coefficient -0.037), so the two registered forecasting primaries, which required a positive coefficient, failed. The data do not separate Ψ_H from the summed response: at a one-hour horizon the estimated responses rarely change sign (median $\Psi_H/|S_H| = 1.00$), the elasticity is the same in contrarian and momentum markets (0.59 and 0.60), and $S = \kappa/M$ itself has elasticity 0.46.

Reading. The certificate measures how close a market is to instability, and in these data it carries information about the coming day's turbulence that the standard measure of self-excitation does not. It does not tell where a crash will land, and the size of the estimated response, though it scales with losses, adds nothing in the predicted direction once recent volatility is known.

7.5 Real agents

H1 is prospective and uses participant-level data. Hyperliquid, a perpetual-futures exchange whose order book runs on a public blockchain, publishes every trade with the buyer's and the seller's wallet address. H1 was registered before any of its data existed; a collector records every trade in 30 markets, ranked 11 to 40 by volume when collection started, and the analysis opens once at least 20 markets have 30 days of near-complete recording. For the ten largest accounts in each market it estimates each account's memory, the persistence of its position, and its response to the displacement, and asks whether two accounts have distinguishable memories in at least 75% of markets, and whether prices and anonymous flow fail to separate memories where the accounts' own records succeed. It is the first test of the identification theory against real agents rather than proxies. Its limits were stated at registration: addresses are accounts, not firms or algorithms, and positions are reconstructed only from the start of collection.

8 Robustness (exploratory)

The analyses in this section were chosen after the registered results were known and are not confirmatory. They use the committed data rows of S1 and R1.

Table 3: Exploratory robustness of S1's variance forecast and R1's size elasticity

S1: margin forecasts next-day variance			R1: elasticity of adverse move		
Variant	coef.	t	Variant	elasticity	s.e.
Registered	-1.57	4.82	Registered	0.59	0.09
Clustered by week	-1.57	3.75	With volatility controls	0.29	0.15
Clustered by market	-1.57	5.10	2025 only	0.97	0.14
Training share 50%	-2.69	4.39	2026 only	0.08	0.23
Training share 70%	-1.51	4.38	Impact $\hat{\kappa}$ alone	0.70	0.10
Winsorized at 1/99%	-1.74	5.52	$\log S$, shock control	0.46	0.12
2025 only	-1.27	2.32	$\log S$, Pilot 7's control	-0.08	0.04
2026 only	-2.22	4.15			
Without October 2025	-0.95	4.33			
High-volatility markets	-1.76	4.84			
Low-volatility markets	-1.43	2.90			
Leave one market out (t range)		4.40–5.37	Leave one market out (range)	0.55–0.63	

Left: S1's pooled out-of-sample forecast of next-day $\log(1 + \text{RV})$ from the winsorized 20-day margin, Clark–West t with differentials averaged within the clustering unit (dates unless stated). Right: R1's 2SLS elasticity of next-day $\log(1 + \text{MAE})$, instrumented by the lagged-window estimate, standard errors clustered by date. Pilot 7's control multiplies the flow shock by the estimated impact.

Table 3 and Figure 3 give two different pictures. S1's variance forecast is stable: the t-statistic stays between 2.32 and 5.52 across clustering units, training shares, winsorization, calendar years, the exclusion of October 2025, volatility halves, and the removal of any single market. R1's elasticity is not. It was 0.97 in 2025 and 0.08 in 2026, it fell to 0.29 with recent volatility in the regression, and the estimated impact $\hat{\kappa}$ alone, the first tap of the response, had elasticity 0.70: the registered result mostly says that days of higher estimated impact are followed by larger moves. The same rows also settle the sign of Pilot 7: with the control Pilot 7 used, which multiplies the shock by the estimated impact, the elasticity of S is -0.08; with the shock alone it is 0.46. The negative sign was a property of that specification, not of contrarian agents.

9 Discussion

For supervisors. The results separate two questions that are often run together. Whether a market is stable can be certified from public data, at a cost set by its distance from instability, without identifying any agent; who is trading cannot be learned from market

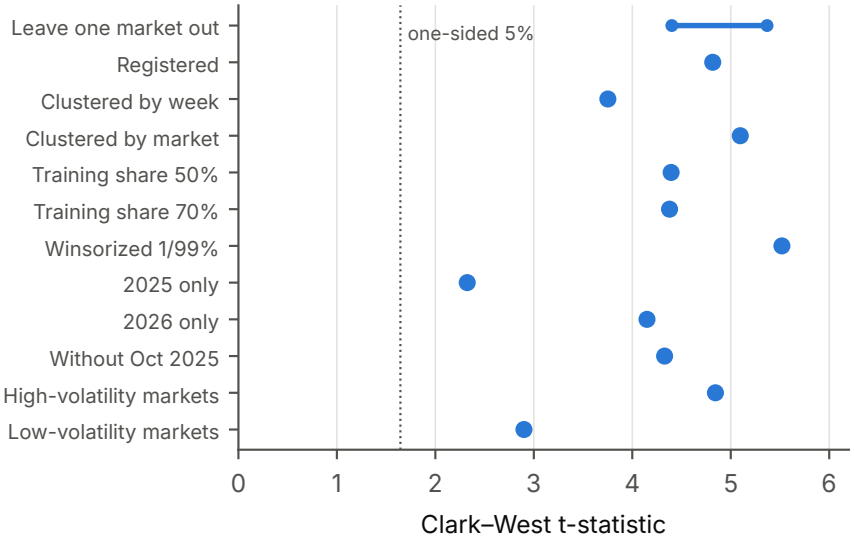


Figure 3: Exploratory robustness of S1's registered variance forecast: Clark–West t-statistics under alternative clustering, training shares, winsorization, subperiods and subsamples, and the range when any one market is removed.

data at any comparable cost. Three practical implications follow. First, a supervisor who cares about stability should estimate the margin, with the pooled estimator and a window matched to the drift of the market, and treat identification as a separate task with its own justification. Second, disclosure has a precise role in certification: it must bound the memory of the largest participants, because without such a bound no passive certificate exists (Theorem 9). Requirements that reveal identities without bounding memory do not help stability supervision. Third, the margin should not be read as a crash indicator. In our data it forecast turbulence and did not rank damage on the worst days.

For research. The tests used one venue and one asset class, and every certified market was far from the boundary. Two open questions matter most. The first is whether the margin falls before genuine near-boundary episodes; this needs events that the present sample does not contain, or a market designed to be near the boundary. The second is whether real agents' memories are distinct and invisible to prices, which H1 answers with participant-level data. On the theory side, the uniform result for geometric memory and for the pooled estimator's nuisance parameters remain open, as does the gap between the moving-average displacement and the distance from fundamental value.

Limits. The model is linear and its agents respond to one state. Real algorithms use many signals, change their parameters, and trade across venues. The companion thesis

shows that the stability bound survives nonlinear rules and time-varying agents given a bound on their total sensitivity, but our estimators and tests use the linear form. The exogeneity of contemporaneous flow to price noise is an assumption of the pooled estimator that five-minute data can violate.

10 Conclusion

A supervisor with public data can learn how close a market is to feedback instability, at a cost set by that distance, without learning who trades in it; learning who trades costs polynomially more and does not get cheaper by watching. The one input prices cannot supply is a bound on how slowly the largest participants trade. In crypto perpetual futures the certificate behaves as the theory says for stability and forecasts the coming day's turbulence beyond the standard measure of self-excitation; it does not forecast crashes. Every one of these statements rests on tests whose design was public before their data were.

Acknowledgements and disclosure. Code, proofs and text were drafted with the assistance of Claude (Anthropic), an AI system; the authors checked and are responsible for all content. Arithmetic steps of the proofs are checked in Lean 4 with Mathlib (modules `ThesisPassiveMulti`, `ThesisHidden`, `ThesisPooled` and companions). Specifications, code, results and the checks that regenerate every number in this paper are in the public repository https://github.com/Pratikn03/Fianince.

A Proofs

Theorem 5. (i) Both economies are $x_{t+1} = \rho_j x_t + e_{t+1}$ with $e_{t+1} \sim N(0, \kappa^2 s^2)$, $\rho_0 = 1-\mu$, $\rho_1 = 1+\mu$. The divergence is $\frac{1}{2}(2\mu)^2 \sum_{t<T} v_t$ with $v_0 = 0$, $v_{t+1} = \rho_0^2 v_t + 1$, so $v_t \le 1/(1-\rho_0^2) \le 1/\mu$ and the divergence is at most $2T\mu$. By Pinsker's inequality the total variation distance is at most $(T\mu)^{1/2}$, while a procedure with error at most α under both separates them with total variation at least $1-2\alpha$. (ii) For fixed $|\rho| < 1$, $\sqrt{T}(\hat\rho - \rho) \to N(0, 1-\rho^2)$ (Mann and Wald, 1943) and $1-\rho^2 = M(2-M)$. Along $\rho_T = 1 - M_T$ with $M_T \to 0$, $TM_T \to \infty$, Phillips and Magdalinos (2007) give $(T/2M_T)^{1/2}(\hat\rho - \rho_T) \to N(0,1)$; since $\hat M / M_T \to 1$ in probability, the studentized interval has asymptotic coverage $1-\alpha$ and excludes zero once $M > z_{1-\alpha/2}(2M/T)^{1/2}$.

Theorem 6. (i) g is real on $[0,1]$, equals one at zero and has no zero there, so $g(1) > 0$. Both economies have finite memory, since any real ϕ arises from $k_i = \phi_i/\kappa$, $i \ge 1$, and

$a = \phi_0 - \kappa k_0$. The divergence is $\sum_{t<T} \mathbb{E}_+[((f_+ - f_-)(\mathcal{L})x_{t+1})^2]/(2\sigma^2)$, and $(f_+ - f_-)(z) = (\rho_- - \rho_+)zg(z)$, so it equals $(2\mu/g(1))^2 \sum_{t<T} \mathbb{E}_+ y_t^2/(2\sigma^2)$ with $y_t = g(\mathcal{L})x_t$. Under f_+, $y_{t+1} = \rho_+ y_t + e_{t+1}$ from $y_0 = 0$, so $\mathbb{E}_+ y_t^2 \le \sigma^2/(1-\rho_+^2) \le \sigma^2 g(1)/\mu$, and the divergence is at most $2T\mu/g(1)$. Pinsker's inequality finishes.

(ii) The regression is an invertible reparametrization of the stationary autoregression of x_{t+1} on $x_t, \dots, x_{t-p}$, so by Mann and Wald (1943) the coefficients are asymptotically normal with covariance $\sigma^2\Gamma^{-1}$, and the entry for x_t is $1/\tau^2$, with τ^2 the residual variance of x_t after projection on the lagged differences. Each such residual is $h(\mathcal{L})x_t$ with h a polynomial of degree at most p and $h(1) = 1$. With $\delta = 1 - \rho$, $G = \max_{|z|=1} |g|$ and $|1 - \rho e^{-i\lambda}|^2 \le \delta^2 + \lambda^2$,

$$\tau^2 \ge \frac{\sigma^2}{2\pi G^2} \min_h \int_{-\pi}^{\pi} \frac{|h(e^{-i\lambda})|^2}{\delta^2 + \lambda^2}\, d\lambda.$$

Let $H(\lambda) = h(e^{-i\lambda})$ and I the integral. Since $\delta^2 + \lambda^2 \le 1 + \pi^2$, $\int |H|^2 \le (1+\pi^2)I$; by Cauchy–Schwarz $\max |H|^2 \le (p+1) \int |H|^2/(2\pi)$, and by Bernstein's inequality $\max |H'| \le p \max |H|$, so $\max |H'| \le pB\sqrt{I}$ with $B^2 = (p+1)(1+\pi^2)/(2\pi)$. With $r = \min(\pi, 1/(2pB\sqrt{I}))$, $|H| \ge 1/2$ on $|\lambda| \le r$, so $I \ge \arctan(r/\delta)/(2\delta)$. If $r \ge \delta$ this is at least $\pi/(8\delta)$; otherwise $r = 1/(2pB\sqrt{I}) < \delta$ and $I > 1/(4p^2B^2\delta^2)$. Hence $I \ge c/\delta$ with $c = 1/(4p^2B^2)$, $\tau^2 \ge \sigma^2 c/(2\pi G^2 \delta)$, and with $\delta = M/g(1)$, $V = \sigma^2/\tau^2 \le 4p^2(p+1)(1+\pi^2)G^2 M/g(1)$. Finally $g(z) = \prod_j (1 - z/r_j)$ with $|r_j| \ge 1 + \gamma$ gives $G \le 2^p$ and $g(1) \ge (\gamma/(1+\gamma))^p$.

Theorem 7. (i) With one cohort of memory rate zero, $f_{t+1} = qx_t + w_{t+1}$ and $x_{t+1} = ax_t + \kappa f_{t+1} + \varepsilon_{t+1}$. The conditional density of (f_{t+1}, x_{t+1}) factorizes into the flow equation and the price equation given the flow, and the second is the same under both economies, so the divergence is $\sum_{t<T} \mathbb{E}_+(q_+ - q_-)^2 x_t^2/(2s_w^2)$ with $|q_+ - q_-| = 2\mu/\kappa$. Under the safe economy $x_{t+1} = (1-\mu)x_t + \kappa w_{t+1} + \varepsilon_{t+1}$, so $\mathbb{E}_+ x_t^2 \le \sigma^2/\mu$ with $\sigma^2 = \kappa^2 s_w^2 + s_\varepsilon^2$, and the divergence is at most $2T\mu(1+R)$. (ii) Write $\hat{M} - M = -(\hat{\beta} + \hat{\kappa}\Phi + M) - \hat{\kappa}(\hat{\Phi} - \Phi)$. Both regressions are equations of a stationary Gaussian vector autoregression whose regressors are uncorrelated with their errors, so the estimators are jointly asymptotically normal and $\hat{\kappa} \to \kappa$. In the flow equation, reparametrized with x_t and p lagged differences, the coefficient on x_t is Φ, so $\hat{\Phi}$ has asymptotic variance s_w^2/τ^2 with τ^2 as above, and $\kappa^2 s_w^2/\tau^2 \le 2\pi G^2\delta/c$. In the price equation, with $\tilde{w}_{t+1} = f_{t+1} - \Phi x_t$, the regression on (f_{t+1}, x_t) is a reparametrization of that on $(\tilde{w}_{t+1}, x_t)$ in which the coefficient on x_t is $-M$; since $\tilde{w}_{t+1}$ lies in the span of w_{t+1} and the lagged differences and w_{t+1} is independent of them and of x_t, the residual variance of x_t after projection on $\tilde{w}_{t+1}$ is at least τ^2, and $\hat{\beta} + \hat{\kappa}\Phi + M$ has asymptotic variance at most $s_\varepsilon^2/\tau^2 \le 2\pi G^2\delta/c$. By $(u+v)^2 \le 2u^2 + 2v^2$, $V_{\text{pool}} \le 8\pi G^2\delta/c = 4 \cdot 2\pi G^2 M/(c\, g(1)) \le 4C_{p,\gamma} M$.

Theorems 9, 10 and 12. The proofs in the text are complete up to the cited limit theory. For Theorem 9, under $\mathcal{E}_0$, $\mathbb{E}_0 x_s^2 \le \sigma^2/(1-(1-c)^2) \le \sigma^2/c$, and $\sum_{t<T}(t+1)^2 \le T^3$; Pinsker's inequality gives TV $\le (\mathrm{KL}/2)^{1/2}$, which tends to zero as $r \uparrow 1$ for fixed T, and a procedure with error at most α under both economies needs $2(1-2\alpha)^2 \le \mathrm{KL}$. The arithmetic steps of all five proofs are checked in Lean 4 (Appendix B).

B Registration record

Table 4: Registration of the tests reported in Section 7

Test	Specification SHA-256 (first 12)	Registration commit	Opened
C2	`aca63504f2ed`	`4a20531`	2026-10-24
C3	`9a4548277fb1`	`f8916b6`	2026-10-25
F1	`f6320484b223`	`b2e78b6`	2026-10-25
S1	`5dc805bb820c`	`b43f5b0`	2026-10-25
R1	`c597bbd7b1a4`	`68bbb75`	2026-10-25
H1	`833b1993b2f2`	`a7a00df`	opens from 2026-12-10

Specifications are in `thesis/checks/prereg/` of the repository, with their hashes and counters. Each test's code verifies the specification hash and the hashes of every pipeline file before running and increments the counter on opening. F1's addendum A1, which restricted the universe filter to ASCII symbol names after a symbol written in Chinese characters stopped the first run before any data were downloaded, was registered in a commit of its own before the run resumed.

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Development revision and validation status: 29 October 2026

This is a working draft for author and specialist review. The revision corrects hidden-state drift accounting, distinguishes cumulative displacement from peak mark-to-market loss, and restricts claims to their stated model and observation contracts. It does not declare every theorem independently verified or every research objective complete.

Returned execution record. The author returned a completed full-mode notebook archive using source commit `19805d7e`, seed 20261029, Python 3.13.18 and NumPy 2.1.3. All 10 exported-file hashes and the four analytical source modules were checked. The run completed E1, E2, E3, E4 and E6; its log reports 18 targeted tests passing. This is an execution reproduction of the same seeded code, not independent scientific validation. No real-market CSV was provided, and the GPU, known-source-feedback and prospective studies were not run.

What the checks establish. E1 cross-checks the corrected drift identity against a resolvent and exercises a sufficient invariant-box guard. E2 demonstrates filter-induced persistence under martingale nulls. E3 compares known alternatives with equal aggregate risk; it is an illustrative witness, not a uniform rate theorem. E4 matches baseline capacity on training episodes and reports test capacity differences; the guard and volatility control both have zero observed budget breaches in the three bounded-shock synthetic regimes. This does not establish universal guard superiority or economic welfare.

Coverage and the observation contract. E6 uses 1,000 replications, 32 reset responses and 512 taps. The nominal 95% interval assumes independent Gaussian tap errors and a valid memory envelope. The intervals in the table measure Monte Carlo uncertainty about coverage, not financial risk.

Setting	Coverage	MC 95% interval	Contract holds
Independent Gaussian	94.6%	[93.0, 95.8]	yes
Correlated tap errors	48.3%	[45.2, 51.4]	no
Heavy-tailed readings	95.1%	[93.6, 96.3]	no
Biased readings	25.6%	[23.0, 28.4]	no
Unseen slow memory	0.0%	[0.0, 0.4]	no

Failures under correlated errors, biased readings and a false memory floor expose the contract's limits. The heavy-tail outcome in this configuration is not a general robustness result. All outcomes, including negative ones, are retained.

Remaining obligations. Independent review of the load-bearing identification and certification proofs, continuous numerical extrema, economic primitives, the public-observation bridge and policy counterfactuals remains necessary. The current two-paper name-map audit finds 54 of 57 formal environments with existing declaration mappings, three unmapped statements and seven revised statements requiring semantic review. A label mapping is not a new Lean proof. The experiment design and full results are retained under `studies/economic_review_20261029/`.

Economic interpretation and returned decision experiment

What this revision adds

The information question is distinct from choosing a preferred amount of risk. For a specified financial decision, the relevant issue is which observations support an exposure constraint, what remains unidentified, and whether obtaining additional information changes the attainable decision after costs. This does not introduce portfolio choice or control without full identification as new ideas. The general identification-versus-decision-value result still requires independent proof and a comparison with the closest existing results.

Time conventions and the institution

The new application is a price-taking, unlevered own-account investor, not a venue-wide supervisor. It begins with cash, buys once, holds a fixed position for 60 one-minute model periods, and closes fully. Round-trip linear and quadratic charges enter cash and wealth. The account cannot change the market's feedback, and the experiment does not measure customer liquidity or welfare. This is separate from the paper's dealer/venue admission contracts.

Quantity	Interpretation
Sampling interval	One minute in the new synthetic account; other historical studies retain their stated sampling clocks.
Historical lookback	Often six hours in the existing forecasting studies; not a holding period.
Forecast horizon	Often the next hour's adverse excursion; not an executed round-trip trade.
New holding period	Exactly 60 model minutes, with explicit entry and exit charges.
Other theory horizons	The existing finite-horizon theorems retain their declared H; no universal holding period is imposed.

Information and decision design

The new comparison uses controlled independent reset observations of 12 impulse taps, a known plant and reading variance, and a declared finite candidate class. It does not treat public market trading as a reset experiment. Common training and validation paths supply 768 and 384 additional reset episodes per repetition; 512 further paths evaluate the selected actions. These are in addition to the reported tap-reading budget, not evidence of learning from 32 observations in total. Verified disclosure restricts the minimum-memory rate to an audited band. The oracle knows the true response, not future shocks. All certificate arms use the same training objective.

The account selects a position by empirical expected logarithmic terminal wealth, subject to its information-supported exposure restriction. This is a finite-horizon application of logarithmic utility, not a newly proved Kelly growth-optimality theorem. Equal response bounds alone do not imply equal expected returns or identical utility-optimal portfolios.

The new account has initial wealth of 1,000 dollars and an initial price of 100 dollars; its position grid runs from zero to 9.9 units. Its round-trip dollar charge is $c(I) = 0.002I + 0.0002I^2$. For bounded external flow, the constraint adds this charge to position times the largest admissible absolute price response over the holding horizon. The resulting bound concerns loss from initial wealth at each date. It is not peak-to-trough drawdown, protection against arbitrary unbounded shocks, or a rolling deployment guarantee.

Returned execution and proof scope

The returned full-mode archive uses seed 20261029. All 35 manifest entries and eight core source files were checked. The run contains 20 independent repetitions per regime, not 16,800 independent experiments: the 16,800 policy rows reuse tapes across policies, budgets

and observation counts. Twelve targeted checks passed. Eight conditional decision-logic lemmas compiled with Lean 4.19.0 and their axiom printouts report no axioms. This is a seeded execution reproduction. It neither completes the older manuscript's Lean coverage nor formalizes the Gaussian confidence argument, market dynamics, general minimax separation, portfolio optimality or welfare.

The displayed slice uses 32 reset episodes and a one-dollar loss budget from initial wealth of 1,000 dollars. It is an illustrative presentation of an exploratory grid, not a primary selected before the results were seen. All regimes, budgets, policies and negative outcomes remain in the result archive. The inputs and complete outputs are retained with the verification record in the new information-decision study directory.

Aggregate information and unresolved cohorts: returned simulation

Regime	Unique candidate (%)	Certified units	Oracle units
No feedback	0.0	9.405	9.900
Moderate feedback	5.0	5.000	5.000
Near boundary	0.0	1.200	1.200

Here "unique" means one remaining candidate in the declared finite class, not identification of arbitrary real-world algorithms. In the moderate and near-boundary configurations, the aggregate certificate reaches the oracle's grid capacity in this slice even though the candidate decomposition is usually unresolved. This is an illustrative decision comparison; it does not prove a uniform rate separation for continuous classes.

Disclosure cost and the negative result

Paid minus aggregate	Mean (bp)	Lower (bp)	Upper (bp)
No feedback	-0.100	-0.100	-0.100
Moderate feedback	-0.126	-0.126	-0.125
Near boundary	-0.100	-0.100	-0.100

The intervals are exploratory percentile bootstraps over repetition-level paired means, not simultaneous intervals across the grid. In this slice the unpaid disclosure does not expand capacity, while the assumed one-cent charge reduces net log wealth. This run therefore does not establish a positive net economic value of disclosure. The price is hypothetical, not a calibrated reporting cost. Stronger disclosure need not be worth purchasing when aggregate information already supports the relevant decision.

Assumption failures and calibration diagnostics

The table reports finite-sample frequencies for all seven regimes, combining the three reading budgets. Each band outcome is counted once per repetition and reading budget (60 outcomes per regime); repeating the same outcome across four financial loss budgets would not create further observations. The three reading budgets within a repetition are not an independent replication.

Regime	Contract valid	Tap-band hit (%)	Empty set (%)
No feedback	Yes	96.7	3.3
Moderate feedback	Yes	93.3	5.0
Near boundary	Yes	96.7	3.3
Hidden slow memory	No	98.3	11.7
Correlated readings	No	98.3	1.7
Biased readings	No	0.0	100.0
Behaviour change	No	95.0	5.0

The nominal one-decision band coverage is 95 percent under the valid contract; these small, dependent-budget frequencies are diagnostics, not a new coverage theorem. A tap-band hit concerns the observed response, not membership in the finite candidate class or stability of future behaviour. In particular, hidden slow memory falls outside the declared class even when its first 12 taps fit the bands. Under behaviour change the readings describe the old response. Biased readings empty the set throughout this run and force the certificate policy to abstain. Correlated readings use an unverified standard-error model. None of these four regimes retains the stated guarantee; a high observed band hit or zero observed financial breaches does not make an invalid contract valid.

Pending evidence: deliberately empty result cells

The following numerical cells remain empty. The returned archive contained no market CSV, dynamic-inventory experiment, institution-level intervention or external replication. The general theorem and independent novelty review also remain open; a table placeholder is not a theorem statement or a proof.

Pending study	Estimate	Uncertainty
Untouched market-data test beyond order-flow imbalance and depth		
Full changing-inventory, execution and funding comparison		
Dealer/venue intervention with endogenous market response		
Net value under empirically calibrated disclosure costs		
Independent implementation and replication		

Further data cannot by themselves prove a general identification or decision-value theorem. Its exact statement, proof and literature priority require separate review. No publication rating follows from the returned run.

Finite-horizon information, cohort ambiguity, and net inventory-service value

Status and contribution boundary. This appendix supplies a restricted analytic separation and its financial consequence. The proof below is written in full; exact rational and numerical checks accompany it. It has not received independent proof review and is not a Lean formalization. Its priority is unconfirmed. Control without unique identification, task-directed exploration, Gaussian functional estimation, moment matching, and the value of information are established ideas (van Waarde et al., 2020; Wagenmaker et al., 2021; Donoho, 1994; Heinrich and Kahn, 2018; Blackwell, 1953). The candidate extension is their explicit combination for a finite-horizon inventory commitment, with a hard stress budget and the information bill deducted from both capital and profit. It does not close the broader open theorem of the preceding drafts.

The observation contract and the financial institution

Fix $H \geq 4$, $0 \leq a < 1$, $\kappa > 0$ and $0 \leq Q \leq \overline{Q} < (1-a)/\kappa$. There are arbitrarily many finite cohorts, with $q_i \geq 0$, $\sum_i q_i = Q$, and $0 < \eta_i \leq 1$:

$$x_{t+1} = ax_t + \kappa\big(\sum_i q_i y_{i,t} + u_t\big), \qquad y_{i,t+1} = (1-\eta_i)y_{i,t} + \eta_i x_{t+1}.$$

Initial states are reset to zero. Let $h_0, \dots, h_{H-1}$ be the price response to a unit impulse, with $h_0 = \kappa$, and put $G = \sum_{t<H} h_t$. Positivity implies $h_t \geq 0$. No positive lower bound on η_i is imposed.

The observer purchases N independent reset experiments, each lasting H model periods and returning all H taps with independent Gaussian errors of known variance σ^2. Thus the tap means are $\overline{h}_t = h_t + \sigma Z_t/\sqrt{N}$, with independent standard normal Z_t. The same observations are available for identifying the cohorts and for choosing the financial position. One experiment is not one scalar reading: it returns H readings and costs c_H, including the declared operating and time charges. Total information cost is $F = Nc_H$. This observation contract is not an ordinary exchange tape; longer horizons cost both time and readings. Plant parameters, the feedback cap, reset, and stationarity are assumptions. Measurements are acquired ahead of a common future service-start deadline. Cash can earn the common reference rate during acquisition; the model does not include earlier service opportunities. The stated measurement price is exogenous, not a calibrated self-impact cost for actual market probes.

The institution is a price-taking, unlevered inventory-warehousing desk. It buys $I \geq 0$ units at P_0, holds them for H periods, and closes. It earns a prequoted service fee pI *after* settlement. Round-trip execution and carry costs are $bI + dI^2$, where $b \geq 0$ and $d > 0$. For example, $b = b_{\text{trade}} + P_0 r_f H/(365 \cdot 24 \cdot 60)$ for one-minute model periods. The service premium is unavailable as collateral until after settlement. Both trading costs and F must be prefunded. Initial cash is W, the stressed-settlement loss budget is B, and $|u_t| \leq \overline{u}$, with $\overline{u} > 0$. Assume $P_0 > \overline{u}\,\overline{G}$, so the asset price remains positive. This desk does not alter market feedback, depth, or another participant's behaviour. The service fee is exogenous, not an equilibrium price.

For the expected-profit objective, future shocks have conditional mean zero given the observations. Consequently the conditional expected trading price change is zero. The financial objective, including every declared charge, is

$$V(I, F) = g(I) - F, \qquad g(I) = (p - b)I - dI^2.$$

This is expected service profit with a hard stress constraint, not a Kelly growth theorem or a return-prediction result. There are no additional training or validation tapes, no leverage, uncertain fills, or changing inventory.

Lemma 13 (Sharp prior bounds and exact prefunded capacity)**.** *Write*

$$\underline{G} = \kappa \sum_{t<H} a^t, \qquad \overline{G} = \kappa \sum_{t<H} (a + \kappa \overline{Q})^t.$$

Then $\underline{G} \le G \le \overline{G}$, and both endpoints are attained in the declared class. For a gain bound $C \ge G$ and $0 \le F \le \min(B, W)$, define

$$R(v, r) = \frac{2v}{r + \sqrt{r^2 + 4dv}}, \qquad I_u = \frac{(p-b)_+}{2d},$$
$$I(C, F) = \min\{I_u, R(B - F, \overline{u}C + b), R(W - F, P_0 + b)\}.$$

This is the optimal expected-profit position subject to the stated stress reserve and prefunding constraints. The zero-violation no-observation robust optimum is $I_{\mathrm{nd}} = I(\overline{G}, 0)$, not an arbitrarily conservative benchmark. The response-informed oracle is $I_\circ = I(G, 0)$; it knows no future shocks.

Proof. Let $k_j = \sum_i q_i \eta_i (1 - \eta_i)^j$. The normalized price response is the renewal sequence with defective interarrival probabilities $s_1 = a + \kappa k_0$ and $s_j = \kappa k_{j-1}$ for $j \ge 2$. Their total mass is $a + \kappa Q < 1$, with the remaining mass placed at infinity. The cumulative interarrival distribution at integer $j \ge 1$ is

$$a + \kappa \sum_i q_i [1 - (1 - \eta_i)^j].$$

It lies between a and $a + \kappa\overline{Q}$. Coupling interarrivals by the same independent uniforms orders every renewal time and hence the number of renewals before H. Taking expectations proves the two bounds. The lower endpoint is the no-feedback economy; the upper is one cohort with $Q = \overline{Q}$ and $\eta = 1$. This also proves sharpness of the no-observation stress bound.

At date $t \le H$, the price change is $x_t = \sum_{j<t} h_j u_{t-1-j}$, so $|x_t| \le \overline{u} \sum_{j<t} h_j \le \overline{u} G$. All costs paid by that date are at most the total prefunded costs. The constant adverse sequence $u_t = -\overline{u}$ attains the total price loss $I\overline{u}G$ at settlement. Before the service fee is credited, worst loss from initial wealth is therefore exactly $I\overline{u}G + bI + dI^2 + F$. The declared robust reserve requirement is $I\overline{u}C + bI + dI^2 + F \le B$; prefunding requires $P_0 I + bI + dI^2 + F \le W$. These increasing quadratics yield the two roots. Finally g increases up to I_u and decreases after it, proving the optimizer and the two benchmarks. This exact reserve contract is not a peak-to-trough drawdown or a repeated-deployment guarantee. □

Lemma 14 (No data at the same error budget)**.** *Let $I_f = \min\{I_u, R(W, P_0 + b)\}$ and allow a data-free rule to randomize, with probability of a stress-budget violation at most γ in every economy of the class. Its greatest expected service profit is*

$$V_{\mathrm{nd},\gamma} = (1 - \gamma) g(I_{\mathrm{nd}}) + \gamma g(I_f).$$

It is attained by the displayed two-position lottery. This is a theoretical benchmark, not a

recommended trading policy. The corresponding response- informed same-error-budget oracle has value $(1-\gamma)g(I_\circ)+\gamma g(I_f)$.

Proof. Truncating an action at I_f weakly improves profit and lowers risk, so restrict to $0 \le I \le I_f$. At the attainable worst economy with gain $\overline{G}$, every $I > I_{\mathrm{nd}}$ is unsafe unless $I_{\mathrm{nd}} = I_f$, in which case all larger actions are already dominated. Thus a feasible lottery puts at most γ mass above I_{nd}. Below that point profit is at most $g(I_{\mathrm{nd}})$, and above it at most $g(I_f)$. This gives the upper bound. The two-point lottery attains it and is unsafe in any economy with probability at most γ. Replace $\overline{G}$ by the known G for the oracle statement. □

Theorem 15 (Uniform cost-aware finite-horizon decision)**.** *Fix* $0 < \gamma < 1/2$, $z = \Phi^{-1}(1-\gamma)$, $s_N = \sigma\sqrt{H/N}$, *and*

$$Y = \sum_{t<H} \overline{h}_t, \qquad C_N = \min\{\overline{G}, \max\{\underline{G}, Y + z s_N\}\}.$$

Purchase a fixed, affordable N *and take* $I_N = I(C_N, F)$. *The following hold uniformly over the declared positive-cohort class, without dependence on cohort spacing, cohort count, or a memory-rate floor.*

1. *With probability at least* $1-\gamma$ *over measurements, the position satisfies the stress budget for* every *bounded future shock path.*

2. *Put* $r_0 = \overline{u}\underline{G} + b$, $I_{\max} = W/P_0$,

$$L_G = \frac{\overline{u} I_{\max}}{r_0}, \qquad L_F = \frac{1}{\min(r_0, P_0 + b)},$$
$$A = 2z\sigma\sqrt{H}\,(p-b)_+ L_G, \quad D = 1 + (p-b)_+ L_F.$$

   With probability at least $1-2\gamma$,

$$0 \le g(I_\circ) - V(I_N, F) \le A N^{-1/2} + D c_H N.$$

   For sufficiently small $c_H > 0$, *choosing* $N = \lceil (A/(2Dc_H))^{2/3} \rceil$ *is affordable and gives a bound* $3\,2^{-2/3} A^{2/3} (Dc_H)^{1/3} + Dc_H$, *when* $A > 0$. *This is an upper bound from this procedure, not a new minimax optimality claim.*

3. *Let* $I_- = I(G + 2z s_N, F)$. *A sufficient condition for strictly positive expected net information value relative to the strongest no-data rule at the* same *error budget is*

$$\underbrace{(1-2\gamma)g(I_-) - F - V_{\mathrm{nd},\gamma}}_{\mathcal{L}_N} > 0.$$

Precisely, $\mathbb{E}_D[V(I_N, F)] - V_{\mathrm{nd},\gamma} \geq \mathcal{L}_N$. *The conclusion can fail at high data costs or a low service premium.*

Proof. $Y - G$ is Gaussian with variance $H\sigma^2/N$. On $\{Y - G \geq -zs_N\}$, which has probability $1 - \gamma$, $C_N \geq G$; clipping preserves this because G lies between the prior bounds. Lemma 13 proves simultaneous safety for every future bounded path on this single event, including the data-selected position. There is no tail beyond H to estimate, explaining the absence of a memory-floor assumption. This does not certify an infinite horizon.

On $E = \{|Y - G| \leq zs_N\}$, of probability $1 - 2\gamma$, $G \leq C_N \leq G + 2zs_N$. The position function is decreasing in each of C and F, so $I_- \leq I_N \leq I_\circ$ on E. To bound the position difference, implicitly differentiate the stress root: its derivatives are $-\overline{u}I/(\overline{u}C + b + 2dI)$ in C and $-1/(\overline{u}C + b + 2dI)$ in F. Clip this root at $I_{\max}$ without changing the final minimum, since the prefunding root is at most $I_{\max}$. The clipped root is L_G-Lipschitz in C and $1/r_0$-Lipschitz in F. The prefunding root is independent of C and $1/(P_0 + b)$-Lipschitz in F. Minimum with I_u preserves these common Lipschitz bounds. Hence

$$0 \leq I_\circ - I_N \leq 2L_G zs_N + L_F F.$$

On $[0, I_u]$, $0 \leq g' \leq (p - b)_+$. Multiplication and addition of the paid fee give the regret bound. Minimizing $AN^{-1/2} + Dc_H N$ over positive real N gives the stated root and value. Rounding up adds at most Dc_H; affordability follows as $c_H N = O(c_H^{1/3}) \to 0$. The case $A = 0$ has no positive-gain information task and does not need this optimization.

On E, $g(I_N) \geq g(I_-)$, while $g(I_N) \geq 0$ always because $0 \leq I_N \leq I_u$. Thus $\mathbb{E}_D g(I_N) \geq (1 - 2\gamma)g(I_-)$. Subtracting F and the same-error-budget no-data value proves the last assertion. Conditional mean-zero future flows justify the stated expected-profit interpretation; their distribution is not learned from extra tapes. □

Theorem 16 (A two-cohort identification obstruction under the same observations)**.** *Fix* $Q > 0$, $a + \kappa Q < 1$, *and* $0 < \eta_0 < 1$. *For sufficiently small* $\delta > 0$, *compare the two economies*

	normalized weights	*adaptation rates*
A_δ	$(1/2, 1/2)$	$(\eta_0 - \delta, \eta_0 + \delta)$
B_δ	$(1/5, 4/5)$	$(\eta_0 - 2\delta, \eta_0 + \delta/2)$.

Actual weights are Q *times those displayed. For fixed* $H \geq 4$, $D_H(\delta) = \|h^{\mathrm{B}}_{0:H-1} - h^{\mathrm{A}}_{0:H-1}\|_2 = \Theta(\delta^3)$. *The exact minimax error for distinguishing these two* known *alternatives from* N *reset experiments is*

$$e_N = \Phi\left(-\frac{\sqrt{N} D_H(\delta)}{2\sigma}\right).$$

Therefore error at most $\beta < 1/2$ requires $N \geq [2\sigma\Phi^{-1}(1-\beta)/D_H(\delta)]^2 = \Theta(\delta^{-6})$. In particular any recovery of the two normalized weights to sup-norm error less than 0.15 with probability at least $1-\beta$ in both worlds must satisfy this testing requirement. Both worlds contain exactly two distinct rates. Both have identical infinite summed response, but their finite-horizon gains are not asserted identical.

Proof. The two rate distributions have equal mass, mean η_0, and variance δ^2. Their third centered moments are respectively 0 and $-3\delta^3/2$. Thus their kernels k_j agree at $j = 0, 1$ and $k_2^{\mathrm{B}} - k_2^{\mathrm{A}} = -3Q\delta^3/2$. The response recursion gives equal h_0, h_1, h_2 and

$$h_3^{\mathrm{B}} - h_3^{\mathrm{A}} = -3\kappa^2 Q\delta^3/2,$$

which is a lower bound for D_H in absolute value.

For an upper bound put $f_t(\eta) = \eta(1-\eta)^t$. Its third derivative is $3t(t-1)(1-\eta)^{t-2} - \eta t(t-1)(t-2)(1-\eta)^{t-3}$, with absent low-degree terms zero. The sum over $t \geq 0$ of its absolute value is at most $12/\eta^3$. For $\delta \leq \eta_0/4$, all intervening rates are at least $v = \eta_0/2$. Taylor's integral remainder in ℓ_1 is therefore at most $2|\eta - \eta_0|^3/v^3$. The matching first two moments cancel the quadratic Taylor terms. The sum of the two third absolute centered moments is $(1 + 17/10)\delta^3 = 27\delta^3/10$, giving

$$\|k^{\mathrm{B}} - k^{\mathrm{A}}\|_1 \leq \frac{27Q}{5v^3}\delta^3.$$

If $K(z) = \sum k_t z^t$ and $H(z) = \sum h_t z^t$, the recursion gives $H(z) = \kappa/[1 - az - \kappa z K(z)]$. Consequently $H_{\mathrm{B}} - H_{\mathrm{A}} = zH_{\mathrm{A}}H_{\mathrm{B}}(K_{\mathrm{B}} - K_{\mathrm{A}})$. The infinite response sums of both economies equal $S = \kappa/(1 - a - \kappa Q)$. Young's inequality yields $D_H \leq S^2\|k^{\mathrm{B}} - k^{\mathrm{A}}\|_1 = O(\delta^3)$. This proves the two-sided order with constants independent of δ.

For equal-covariance Gaussian laws, the likelihood-ratio test projects the sample mean onto the difference of the two known means, and thresholds at their midpoint. The projected normal law gives the displayed error in each world. Its equal-prior Bayes optimality and symmetry establish minimax optimality. A weight estimator accurate within 0.15 distinguishes the two disjoint normalized-weight balls, proving the identification lower bound. Finite-horizon gains need not be equal: only their difference is of cubic order. The test concerns latent cohort recovery, not identification of the input-output response vector, and does not cover alternative probe designs. □

Corollary 17 (A priced separation)**.** *Fix $c_H > 0$ and an affordable N for which $\mathcal{L}_N > 0$ at the limiting merged economy $\delta = 0$. For all sufficiently small positive δ, the financial rule of Theorem 15 has a strictly positive expected net information value, with its stated safety*

probability. Its observation requirement has no dependence on δ. At this same fixed information price, sufficiently small δ makes cohort identification at error β unaffordable within the same hard fee budget B: $c_H[2\sigma\Phi^{-1}(1-\beta)/D_H(\delta)]^2 > B$. Any procedure limited to that many affordable observations is covered, including one that stops early, since it can be viewed as a function of the full affordable sample. A rational agent may abandon identification and use the no-data rule. There is no assertion that regulation requires identification.

Proof. Apply the two theorems. The first statement uses the strictly positive lower bound on net value, rather than an assumption that more information is automatically beneficial. The second follows from $D_H = \Theta(\delta^3)$. For a neighbourhood of a fixed δ where the strict value inequality holds, continuity preserves it. To claim the inequality for all sufficiently small δ, it suffices that it hold strictly at the limiting merged economy: finite G_H and I_- are continuous there. The numerical example reports both the primary pair and that limiting check. No finite-class grid is used in the uniform decision guarantee. □

What the result does not establish. The direct gain-sum certificate is exactly ordinary Gaussian projection onto the vector of ones. It is not a new estimator and does not beat that benchmark. The latent-cohort lower bound is not a lower bound for learning the observable response vector; comparing it with FIR model-identification results for $\mathcal{H}_\infty$ norms would conflate different tasks (Tu et al., 2018). No optimal estimation rate over all observation designs is claimed. The data cost is not an empirically measured exchange probing cost. An ordinary public-tape bridge, endogenous quotes and depth, incomplete resets, unknown or misspecified noise, and independent replication remain open. The information purchase in the example is predeclared. Its positive value is pointwise at the displayed economy; no prior over unknown economies or globally optimal Bayesian/robust acquisition policy is claimed. The fixed horizon, independent measurement contract, and conditional mean-zero future flows cannot be removed by the simulation. At another horizon the certificate must be recomputed. A rolling sequence needs simultaneous or time-uniform coverage; a one-decision guarantee is insufficient.

Literature priority. The reviewed sources already establish the general principles used here. The explicit dollar-cost/capital formulation and the combined result above are candidate additions to this project, not verified first-in-literature claims. A specialist must check whether they are an immediate corollary of existing task-specific or constrained-decision results before publication.

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Checked synthetic financial illustration

This illustration uses the frozen primary design and the stronger comparison recorded in the design amendment. It is exploratory and was developed after the earlier project runs; it is not independent preregistration or replication. The two-cohort primary has $a = 0.4$, $\kappa = 1$, $Q = 0.4$, rates $0.49, 0.51$, and equal weights. The known class cap is $\overline{Q} = 0.55$; $H = 60$ one-minute model periods, $N = 32$, $\sigma = 0.02$, and $\gamma = 0.025$. Initial wealth is 1,000, price 100, stress budget 1, and flow bound 0.04. The service fee is 0.08 per unit. Execution costs are $0.002I + 0.0002I^2$ and carry is charged at 5 percent annualized. The assumed all-in reset fee is 0.002, so the learning bill is 0.064. These are hypothetical model dollars, not an exchange calibration or returns. The same-error-budget no-data lottery is a theoretical strongest comparator, not a proposed desk practice. The zero-violation comparisons are also shown.

Policy	Mean units	Net expected profit	Unsafe reserve (%)
No data, zero violations	1.306	0.1007	0.00
Best no data, same error budget	1.523	0.1171	2.50
Finite-horizon certificate	4.553	0.2844	2.44
Standard Gaussian projection	4.553	0.2844	2.44
Full-response ellipsoid	4.386	0.2718	0.00
Point estimate (uncertified)	4.601	0.2880	50.41
Exact gain report, fee 0.05	4.669	0.3072	0.00
Response oracle, zero violations	4.914	0.3756	0.00
Response oracle, same error budget	5.041	0.3851	2.50

Profit integrates the future zero-mean flow model and every declared cost; the data-dependent policy values are also integrated over reading noise by one-dimensional Gaussian quadrature. Mean units and empirical reserve violations use 20,000 synthetic reading draws. Oracle and no-data lottery values are analytic. A reserve violation means some bounded future path can exceed the budget, not that a sampled market path actually did so. The 97.5 percent safety statement is over measurement draws for a single future commitment; it is not certainty, market-wide stability, or time-uniform coverage. The point rule buys slightly more expected profit by violating the same risk requirement and is not a valid superiority comparison. The proposed estimator and standard Gaussian projection coincide exactly.

The net information advantage over the strongest same-error-budget no-data rule is 0.1673. The written theorem's sufficient lower bound is 0.1465, strictly positive even after all costs. At the limiting merged economy the lower bound is 0.1465; continuity therefore supports the small-separation corollary. These bounds are evaluated in floating point with a wide positive margin, not claimed interval-arithmetic certificates.

Separation δ	Known-pair resets required	Implied fee bill
0.0800	478	0.955
0.0400	3.96e+04	79.2
0.0200	2.84e+06	5.67e+03
0.0100	1.91e+08	3.83e+05
0.0050	1.26e+10	2.51e+07
0.0025	8.15e+11	1.63e+09

The known-pair test receives all 60 taps, exactly as the financial rule does. Response differences are calculated in rational arithmetic before conversion to floating point. The sample threshold is the Gaussian testing formula, not a measured runtime or an asserted physical trading-capital requirement. For the primary $\delta = 0.01$, distinguishing the two already-known

alternatives requires about 191 million resets at 5 percent error. Even this optimistic information fee is far beyond the unit stress budget. Identifying arbitrary cohorts cannot be easier than this embedded test under the same fixed query contract. Other experimental designs are not covered.

At fixed $N = 32$, the expected net benefit crosses zero at an illustrative reset price of about 0.005825. An exact scalar gain report is worth purchasing only below a fee of about 0.188690 in this example. These are computed break-even prices, not observed reporting costs.

Failures retained. The full output includes all 819 cost, horizon and service-fee sensitivity rows, including negative values and unaffordable information budgets. A low service premium or expensive measurements can make abstaining from data acquisition preferable. Independent checks cover 13 algebra, optimizer, shock-corner, cash-flow and Gaussian-testing properties, including the same-error-budget no-data optimum checked against a separate linear program. An initial low-premium LP check was affected by the solver's absolute objective tolerance; rescaling the objective resolved it, and that failed log is retained. This is not independent mathematical proof review.

Reading/model contract	Gain coverage (%)	Unsafe reserve (%)
Independent Gaussian, valid	97.36	2.63
Correlated taps, wrong variance	71.16	28.84
Correlated taps, correct variance	97.36	2.63
Negative reading bias, invalid	0.00	100.00
Future feedback changes, invalid	0.00	100.00

The valid Gaussian row has finite Monte Carlo variation around its nominal 97.5 percent coverage. Using the correct correlated variance restores the specified measurement contract at the cost of a larger margin. Bias or future parameter change is not repaired by a larger sample under the wrong model. A signed kernel such as $(1, -0.9)$ has sum 0.1 and absolute sum 1.9, so positivity is indispensable to using the plain sum. The same certificate used repeatedly without correction does not retain its one-decision guarantee.

What changed economically. In this model, the desk can choose a larger inventory commitment while paying for information and retaining the same permitted unsafe-action probability as its strongest no-data comparator. This is an example of positive decision value, not evidence that information always pays, that trading strategies must be identified, or that a real venue's liquidity and welfare improve. Realistic observation costs and a public-data bridge remain requirements for an empirical finance claim.

Returned advanced experiment: evidence and limits

Research question and financial institution. Can information about an aggregate market response improve an exposure decision after information, execution and carry costs, while the hidden trading cohorts remain difficult to recover? The new experiment concerns a price-taking, unlevered inventory-warehousing desk. It buys a fixed inventory, holds it for $H = 10, 60, 240$ one-minute *model* periods and closes. A prequoted service premium arrives after settlement. It cannot change market feedback, quotes or other traders. This is a separate application from a dealer or venue that changes admission and thereby changes the market itself. The objective is expected service profit under a stress reserve and cash constraint, not logarithmic portfolio growth or predicted crypto returns.

Completed execution record. The author returned the Deep Research archive with base seed 20261101. All 611 manifest entries, all 288 checkpoint-to-statistics links and all 15 released source files were verified. The 288 generated economies use 24 seed clusters, 4,096 measurement repetitions per economy and N, $N = 8, 32, 128$, three information-price settings, seven measurement rules and a same-error oracle. The 20,736 policy/cost rows share readings and are not independent market experiments. All 27 implemented tests passed. Numeric policy fields reproduce the local same-seed run to about 1.25×10^{-14}. This is cross-environment reproducibility of the same implementation, not independent replication. The returned runtime used Python 3.13.18 and the pinned numerical packages; the research runner took 52.71 seconds, excluding notebook setup.

What Lean checked. Lean 4.19.0 checked seven concrete rational inequality certificates, with no reported axioms. They include a conservative positive-value numerical witness greater than 0.14 hypothetical model dollars for the original primary contract. They do not formalize the Gaussian analytic tail step, market dynamics, general separation theorem, translation from observations to economics or novelty. The earlier positive illustration is retained as a pointwise example. It must not be read as a positive average-value theorem over all economies or prices.

Economic value depends on the information bill. For the standard Gaussian rule within the independent-Gaussian economy design:

Information price	Mean advantage	95% interval	Positive / total
Free	+0.13047	[0.08458, 0.18420]	99 / 144
Baseline	−0.02953	[−0.07022, 0.01310]	59 / 144
Five times baseline	−0.17212	[−0.21856, −0.12715]	26 / 144

The comparator is the analytically optimal no-data rule with the same permitted unsafe-action probability, $\gamma = 0.025$. Values are hypothetical dollars per inventory-service decision. Baseline cost is $N(0.0005 + 0.000025H)$; its dependence on horizon and N is explicit, not

an exchange calibration. The intervals use 5,000 percentile bootstrap resamples of 24 seed-cluster averages. This price-specific analysis was added after the return and is exploratory, not preregistered confirmation or uncertainty about actual market profit. The baseline-price interval crosses zero: this design does not establish a positive average benefit after costs. The high-price interval is negative under the stated simulator sampling scheme.

Protection and profitability are different outcomes. For purchased iid-Gaussian rules in iid-Gaussian economies, the mean simulated unsafe-action frequency was 0.834%, below the 2.5% allowance. Across 6,535 declared-valid purchased-rule checks, Holm correction produced zero rejections. Individual finite-sample frequencies can exceed the allowance; non-rejection is not a proof. An unsafe action means its reserve can breach the budget under a permitted worst-case shock path. It is not an observed liquidation, default or historical crash. The Gaussian and declared-variance rules coincide exactly under iid readings. Their advantage over a full-response ellipsoid is a task-specific comparison of established methods, not a new estimator.

Failures retained and interpreted.

Purchased Gaussian rule: observation/future setting	Mean unsafe action
Independent Gaussian readings, stationary future	0.834%
Correlated readings, iid variance used	14.695%
Downward-biased readings	31.169%
Future feedback changes	61.033%

These are unweighted descriptive averages of dependent configuration rows, including the zero-price setting. They are not pooled market failure probabilities. Under correlated Gaussian readings, a declared-variance rule reduces the mean to 0.409%, and a projected Student-t rule to 1.153%. Protection can improve while the information purchase remains uneconomic. The Student-t guarantee requires independent Gaussian reset totals; good observed performance on t_3 noise does not extend that guarantee. Median of means needs independent resets and an externally valid variance bound. At insufficient N it uses a paid prior-bound fallback. Neither robust interval repairs mean bias or a change in future dynamics. The full design retains 7,489 negative-advantage rows and 1,743 unaffordable purchase rows. Pooling all methods and scenarios into one success percentage would be misleading.

Information acquisition is an open economic decision. The fixed-N protocol buys information when affordable; it does not optimize the purchase. In 32 of 288 generated economies the known service premium is no greater than known linear execution/carry cost. Positive inventory then has no positive expected service surplus before any information is obtained. In the first indexed case, s000_c000 at $N = 32$ and baseline price, no-data and response-informed oracle profit both equal approximately 0.75902, because response information cannot relax the relevant position limit. Paying 0.024 for information leaves

approximately 0.73500. A useful measurement need not change the financial action. A purchase rule should use only information available before purchase, include the option to abstain, and avoid selecting N or a policy after looking at hidden gains or test outcomes. That rule is still to be proved and validated; it has not been added to this returned run.

Observation boundary and contribution status. There are no historical coins, exchange tapes or crash replays in this run. The quote-data stage is **NOT_RUN**; the unit-test quote fixture is synthetic. Controlled resets, known plant/noise inputs, a feedback cap and stationarity are maintained assumptions. The finite-horizon positive-response certificate needs no adaptation-rate floor when every relevant tap is observed under that reset contract. This does not remove the hidden-tail problem for infinite-horizon or ordinary public-tape certification. New observations do not independently verify the earlier passive, capital, AI or equilibrium claims.

The supported interpretation is a conditional financial separation: aggregate response information can support a specified exposure decision without latent cohort recovery, and positive net value requires additional economic conditions. The written general proof still needs independent review and its distinction from existing identification, data-informativity and decision theory needs specialist assessment. No publication-quality rating follows from execution.

Traceability. The immutable experiment source is repository commit `6a319b0db147`. The return audit is in commit `c0c0c010ec00`. The study directory and its relative audit directory are shown separately for readability:

```
studies/advanced_cost_separation_20261029/
validation/returned_colab_20261030/
```

The uploaded archive is:

```
MM_advanced_deep_seed20261101_v1.zip
```

Its SHA256 begins `e35c10f6b276`; the full hash, configuration, source fingerprint and exploratory price/paired audits are retained in that directory. No original return outcome has been overwritten.

Research work register: 30 October 2026

This register is a dated snapshot, not a claim that the project is complete. *Completed* means the named artifact or check exists. *Written, review open* means an argument is present but has not received independent mathematical sign-off. *Pending* means the evidence or proof is still required. Good outcomes from another experiment do not silently close a task.

Completed records to preserve

Record	Verified completion and boundary
C1	Returned archive, source fingerprint and 288 checkpoint links verified. Same-code reproducibility; no independent implementation.
C2	All 27 implementation checks passed. They do not test every theorem in the papers.
C3	Seven numerical Lean certificates passed. The full general theorem is not formalized.
C4	288-economy synthetic audit completed; every negative and unaffordable row retained. No market-data stage ran.
C5	Pointwise positive-value illustration retained; broad price-sensitive results and assumption failures recorded. No universal profitability claim.
C6	Targeted literature comparison and a written restricted separation exist. Originality and complete proof review remain open.

Core work before stronger contribution claims

R1. Establish the precise literature gap. Status: pending specialist review. Compare the exact assumptions, observation experiment, decision target and conclusion with identification theory, data informativity, task-optimal exploration, functional estimation and constrained financial decisions. Separate latent cohort recovery from learning the observable response. Do not claim that control or allocation without complete identification is new. *Completion evidence:* a source-backed comparison table identifying a specific non-overlap, or a clear statement that the result is an application or corollary; resolved reviewer comments. More simulations cannot complete R1.

R2. Independently audit the load-bearing proofs. Status: written, review open. Review the finite-cohort renewal bounds, observation-to-confidence implication, uniformity in hidden memory, fixed-query testing obstruction, all fee/cash constraints and the same-error no-data optimum. Audit the older broad certification, passive-identification and capital claims separately. Check quantifiers, constants, admissible probes, reset assumptions and counterexamples. *Completion evidence:* theorem-by-theorem assumptions and proof register, resolved objections and independent mathematical review. Record each Lean theorem's actual scope; do not substitute seven scalar facts for a general proof.

R3. Make the information purchase economically explicit. Status: pending. Define what the desk knows before buying observations and whether the acquisition objective is robust, Bayesian or another stated criterion. Include abstention and screen cases where the premium cannot cover known costs or information cannot change the constrained position. Select sample size and price using available information, not hidden gains or test outcomes. *Completion evidence:* a precise acquisition rule, a valid decision-value or regret statement, matched-information/cost comparisons, and a frozen new validation design. Reusing this return to design a rule is exploratory.

R4. Justify the financial institution and measurement contract. Status: partial. The fixed inventory, horizon, post-settlement premium and accounting are explicit. The economic source of the premium, operating cost of reset observations, information reliability and acquisition opportunity cost need justification. An own-account investor, warehousing desk, market maker and venue supervisor have different actions and incentives. Keep their claims separate. *Completion evidence:* a coherent institutional mechanism and documented parameter/cost interpretation, with comparative statics. If logarithmic utility is introduced, rederive the objective rather than relabeling expected profit.

R5. Establish an observation bridge or restrict the empirical claim. Status: pending. Ordinary public quotes are not independent resets. Address initial state, fundamental-value proxies, endogenous order flow, changing parameters, noise covariance, sign changes and unobserved tails. A reduced-form VAR score is not automatically the structural gain in the theorem. *Completion evidence:* a justified identification/partial-identification argument under stated conditions, or a forecasting-only interpretation with certification claims removed from that application. Test the conditions rather than treating good forecasts as proof of them.

R6. Run a protected real-data evaluation. Status: NOT_RUN. Document event-level quote provenance, instruments, units, dates, fees, carry and earlier use. Freeze venues/dates and specifications before test access. Use chronological purging, volatility and OFI/depth baselines, prespecified inference units and all failure outcomes. An exposure replay must account for spread, funding, fill/latency assumptions and capital at risk. Distinguish isolated round trips from a portfolio backtest and from market-wide intervention. *Completion evidence:* licensed/reproducible data description, immutable split and code, held-out results after costs, limitations and external replay. The synthetic unit-test fixture cannot complete R6.

R7. State what happens when certification is unavailable. Status: partial diagnostics. The returned bias and future-change cases demonstrate failure. Specify a defensible response such as abstention or a prior-only restriction; determine what verified disclosure could restore the missing assumption and what it costs. Repeated use requires simultaneous

or time-uniform control, not repeated use of a single-decision confidence level. *Completion evidence:* a stated contract/failure-response rule with a valid guarantee or an explicit impossibility boundary, tested under prespecified violations. Larger samples under a wrong mean do not repair the problem.

R8. Obtain independent replication and reconcile claims. Status: pending. Have another implementation reproduce the central calculation and meaningful stress cases. Reconcile abstracts, conclusions, theorem tags, old result tables, duplicate paper layouts, co-author contributions and AI-use statements. Keep the original evidence and limitations visible. Treat internal "proved" tags as statements about written arguments until independent review is recorded. *Completion evidence:* an external replication record, a claim-to-evidence map, resolved discrepancies and co-author approval for circulation.

Extensions that remain ambitions

Endogenous dealer depth and quotes, venue-wide gating, equilibrium welfare, dynamic inventories and AI-adoption externalities are separate extensions. They should enter the main claim only with a proved mechanism and appropriate evidence. The new fixed-position experiment does not establish those outcomes. A strong core contribution need not contain every extension.

Order of work and maintenance rule

Prioritize R1–R3 together: determine what is new, whether it is true, and when the information purchase changes an economically meaningful decision. R4–R6 determine the strength of the finance application. R7–R8 determine defensible use and reproducibility. Professor review can guide and challenge each item; it is not a substitute for producing the missing proof or evidence.

For each update record the date, task ID, changed claim, immutable code/data reference, result and remaining limitation. Mark an item completed only when its named evidence exists. Update this register, its Markdown/JSON source and all PDF variants together. Publication quality remains a judgment about originality, correctness and usefulness, not a score produced by the notebook.